



Morphing-Based Sensitivity Analysis: A Comparative Study of Linear and Non-linear Temporal Transformations in TSC Robustness

Antónia Brito^{1,2} · Duarte Folgado^{3,4} · Carlos Soares^{1,2,3,5} · Moisés Santos^{1,2}

Received: 27 February 2026 / Revised: 15 July 2026 / Accepted: 22 July 2026

© The Author(s), under exclusive licence to Springer Science+Business Media LLC, part of Springer Nature 2026

Abstract

Machine learning models deployed in high-stakes domains such as healthcare must remain both accurate and robust to subtle, semantically meaningful temporal changes between classes. However, existing sensitivity-analysis methods for time series classifiers typically report aggregate robustness without validating whether their underlying metric reflects the position of a classifier's decision boundary, and it is unclear whether this holds equally across different choices of the morphing operator such methods rely on to generate intermediate instances. We propose tsMIST (Time Series Model Sensitivity Test), a framework that morphs borderline pairs of real instances from different classes and summarizes, via two measures, $tsMIST_{Avg}$ and $tsMIST_{Std}$, the point along the morphing path at which a classifier's prediction switches class. We prove formally, and confirm on data with a known decision boundary, the conditions under which $tsMIST_{Avg}$ estimates boundary position, showing that its commonly-assumed reference value of 0.5 reflects source-target symmetry rather than a universal target. We further compare linear interpolation against a path-interpolation (PathI) operator that slides matched features in time via their dynamic-time-warping correspondence rather than blending amplitudes. The results show that linear morphing is structurally unable to recover boundary position when the discriminative signal between classes is temporal rather than amplitude-based, whereas PathI recovers it exactly. On ten real-world medical time series datasets and four classifiers, InceptionTime consistently exhibits the most robust decision boundaries and Catch22 the most fragile; PathI and linear morphing differ systematically in estimated boundary position without a practically meaningful difference in consistency, and PathI more closely preserves the temporal structure of the original signal. We conclude that the choice of morphing operator materially affects sensitivity-analysis outcomes and should be justified and reported rather than assumed, supporting structured, boundary-aware sensitivity analysis as a component of Responsible AI evaluation.

Keywords Time series classification · Model robustness · Dataset morphing · Sensitivity analysis

Editors: Gianvito Pio, Jurica Levatić, Nikola Simidjievski.

Extended author information available on the last page of the article

1 Introduction

In critical domains such as healthcare, machine learning models must exhibit not only high accuracy but also robustness against subtle input perturbations (Balendran et al., 2025). Traditional evaluation measures often overlook model sensitivity to semantically meaningful variations, potentially masking vulnerabilities in real-world deployment (Sehwag et al., 2019). This gap is particularly relevant in time series classification tasks, where gradual temporal pattern changes – such as variations in medical signals (Mohammadi Foumani et al., 2024) – may lead to critical diagnostic errors. A classifier that distinguishes a normal from a pathological electrocardiogram, for instance, must remain reliable not only on prototypical examples but across the continuum of borderline morphologies that arise from physiological variability, sensor noise, or disease progression; a decision boundary that flips under a clinically negligible deformation, or that resists a clinically meaningful one, is a latent source of diagnostic error that aggregate accuracy does not expose.

Recent studies address model robustness through three main perspectives: performance analysis under adversarial scenarios, spatial mapping of model behavior, and resilience to distributional shifts. Work by Inácio et al. (2025) and Drapal et al. (2024) emphasizes identifying failure-prone regions, while (Sehwag et al., 2019) highlights risks from adversarial attacks. However, few methods explore structured perturbations that preserve temporal semantics, limiting realistic model evaluation. *Time series morphing* offers such a perturbation: rather than adding a bounded, worst-case perturbation to a single instance as in adversarial attacks, it generates a sequence of intermediate series that gradually transform a source instance of one class into a target instance of another, tracing a semantically meaningful path across the decision boundary. Observing where along this path a classifier changes its prediction turns an otherwise opaque boundary into a directly measurable quantity.

This paper proposes tsMIST (Time Series Model Sensitivity Test), a sensitivity analysis framework built on time series morphing. Inspired by *tsMorph* (Santos et al., 2024), originally developed for forecasting, tsMIST adapts morphing to classification by quantifying a model's resistance to gradual inter-class transformations. For each borderline pair of instances from different classes, the method computes the *switchThreshold*, the normalized morphing point at which the prediction first deviates from the source class, and summarizes it across pairs into two measures, $tsMIST_{Avg}$ and $tsMIST_{Std}$. Crucially, we do not take the interpretation of these measures for granted: we establish, both formally and through a controlled synthetic validation, the conditions under which $tsMIST_{Avg}$ recovers the true position of the decision boundary along the morphing path, and show that its frequently-assumed reference value of 0.5 is a consequence of source–target symmetry rather than a universal ideal.

Experiments encompass synthetic data with sine wave patterns and variable noise, plus 10 real-world medical signal datasets from the Aeon Time Series Classification archive. Four classification algorithms were evaluated: an LSTM network, a Random Forest with Catch22 features (Lubba et al., 2019), a Ridge Classifier with randomized kernels from Rocket (Dempster et al., 2020), and the InceptionTime classifier (Ismail Fawaz et al., 2020). Results demonstrate that tsMIST identifies vulnerabilities undetected by conventional measures, such as unstable decision boundaries in feature-based models. The code is publicly available at <https://github.com/Nia3324/tsMIST> for reproducibility and further experimentation.

This work extends a previous conference version (Brito et al., 2025) along three axes. First, we replace the single morphing operator of the original framework with a comparison of two complementary operators, linear interpolation and a path-interpolation scheme that interpolates the dynamic time warping correspondence between source and target, allowing discriminative features to slide continuously in time rather than be superimposed. Second, and central to this extension, we introduce a boundary-position validation methodology: rather than assuming that the `switchThreshold` reflects decision-boundary position, we test this claim directly against a classifier whose boundary is known by construction. Third, we audit the quality of the generated intermediate series using `pyMDMA` (Façoco et al., 2025), a framework for auditing real and synthetic data. The contribution of this work is thus not a new classification algorithm, nor either morphing operator in isolation, but an evaluation methodology whose central metric is validated rather than assumed.

This work aims to: (1) propose an interpretable and *validated* robustness measure for temporal classification, together with the conditions under which its interpretation holds; and (2) demonstrate the framework in controlled and real-world scenarios, comparing morphing operators on their ability to expose boundary-level fragility. This approach reinforces Responsible AI practices by making the trade-off between accuracy and boundary stability explicit and measurable.

The paper is organized as follows: Sect. 2 reviews related work; Sect. 3 details the `tsMIST` methodology; Sect. 4 describes the experimental setup; Sect. 5 presents results; and Sect. 6 discusses implications and future directions.

2 Related Work

This section reviews literature across three domains. We first examine Time Series Classification methods to establish the algorithmic context. We then analyze Model Robustness approaches, including stress testing, spatial behavior mapping, and distributional shift analysis. These works motivate sensitivity evaluation beyond accuracy. Finally, we review Dataset Morphing techniques that generate controlled perturbations while preserving semantic structure. `tsMIST` combines these three areas to assess classifier stability under temporal deformations.

2.1 Time Series Classification

Time Series Classification (TSC) methods are often organized according to the type of transformation they apply to the data (Bagnall et al., 2017; Middlehurst et al., 2024), including convolutional approaches (Dempster et al., 2020), deep learning models (Ismail Fawaz et al., 2019; Ismail-Fawaz et al., 2022), dictionary-based methods (Schäfer, 2015), distance-based techniques (Lines & Bagnall, 2015), and feature-based classifiers (Barandas et al., 2020; Lubba et al., 2019). In healthcare, TSC techniques are used to interpret physiological signals, such as electrocardiograms, and support clinical decision-making (Neves et al., 2021). Industrial sectors use TSC for predictive maintenance by monitoring sensor data to foresee equipment failures, thereby optimizing operational efficiency (Resende et al., 2021; Sousa et al., 2025). The versatility of TSC across these fields underscores its significance in extracting meaningful insights from temporal data.

2.2 Model Robustness

Approaches to model robustness relevant to auditing can be categorized into three primary dimensions: performance-based stress analysis, spatial characterization of model behavior, and resilience to distributional shifts and adversarial interference.

Moving beyond aggregate accuracy, recent methodologies focus on mapping model behavior under specific failure conditions. Techniques have been developed to define localized performance regions that delineate reliability zones (Drapal et al., 2024) and to estimate the likelihood of severe forecasting errors through meta-learning (Inácio et al., 2025). These stress-testing frameworks have expanded to include evaluations under unexpected temporal disruptions (Garred et al., 2025) and benchmarks designed to assess the deep semantic understanding of time series models (Cai et al., 2024), ensuring that high performance implies genuine pattern recognition rather than memorization.

Complementary to performance metrics, spatial mapping techniques project complex input distributions into lower-dimensional instance spaces. By visualizing the relationship between data characteristics and classification difficulty, these methods expose structural weaknesses and hard-to-learn regions that remain hidden in standard evaluation reports (Muñoz et al., 2018).

The susceptibility of models to distributional shifts remains a critical auditing concern. Research indicates that standard detectors are often insufficient against adversarial out-of-distribution inputs (Kielhöfer et al., 2025; Schwag et al., 2019). Furthermore, safety-critical systems face risks from increasingly sophisticated vectors, for example from complex temporal attack strategies (Kurniawan et al., 2026). Consequently, robust auditing must address these risks holistically. This includes aligning robustness objectives with fairness and bias mitigation (Lee et al., 2021), ensuring the stability of model explanations under input variations (Nguyen et al., 2024), and employing global sensitivity analysis (Lo Piano et al., 2023) to quantify prediction stability across the entire machine learning lifecycle (Balendran et al., 2025).

tsMIST is adjacent to two related evaluation paradigms, from which it is worth distinguishing explicitly. The first is confidence calibration (Bellinger et al., 2021; Dong et al., 2025), particularly relevant in imbalanced settings where post-hoc calibration is common practice even in safety-critical domains such as medical imaging (Rousseau et al., 2025); calibration evaluates the fidelity of a model's predicted probability to the true outcome frequency, a pointwise property of a single prediction. The second is explainable AI, and explanation-stability metrics in particular (Nguyen et al., 2024), which ask whether a model's post-hoc explanation remains consistent under input perturbation, together with counterfactual testing approaches (Kjærnlí et al., 2024) that probe forecasting models by constructing alternative inputs and observing the resulting change in output. tsMIST shares the counterfactual spirit of the latter but targets a different object than either paradigm: rather than a pointwise probability estimate or a local explanation's consistency under an arbitrary perturbation, it constructs a semantically meaningful path between two *real* instances of different classes and measures where along that path the classifier's decision itself changes, a path-dependent property that neither calibration nor explanation-stability metrics are designed to capture. Unlike calibration metrics, tsMIST also does not require the classifier to expose a probabilistic

output. A direct empirical comparison against calibration and explanation-stability metrics, where both are computable for the same classifiers, is an open direction we leave to future work.

2.3 Dataset Morphing

Dataset morphing has been established as a systematic method to probe model behavior through controlled distributional changes, supporting auditability by exposing gradual failure modes. Methodologies in this domain generally fall into two categories: direct interpolation of data instances and generation via high-dimensional latent spaces.

One class of methods focuses on generating smooth trajectories between real data points to examine performance transitions. Originally applied to collaborative filtering to analyze meta-feature variations (Correia et al., 2019), this concept was extended to time series forecasting with frameworks like tsMorph, which create semi-synthetic datasets for structured testing (Santos et al., 2024). Recent advancements have further refined this approach by combining genetic algorithms with interpolation to actively search for failure boundaries and out-of-distribution samples (Kjærnlí et al., 2024).

A second group applies morphing within continuous latent representations to ensure semantic consistency. While earlier approaches utilized implicit neural representations (Schardong et al., 2024) and normalizing flows to preserve likelihood during transformations (Golling et al., 2023), state-of-the-art methods now leverage diffusion models to generate richer, correspondence-aware morphing trajectories (Liu et al., 2026; Zhang, 2024).

Specific to the temporal domain, ensuring the plausibility of generated samples is critical. To prevent the creation of incoherent artifacts during interpolation, recent techniques enforce constraints based on Dynamic Time Warping (Belkhouja et al., 2022) and temporal smoothness criteria (Pialla et al., 2025). These constraints are essential for evaluating whether models, such as the lightweight LITE architecture (Ismail-Fawaz et al., 2025), maintain robustness under realistic deformations rather than just arbitrary noise (Liu et al., 2025).

Taken together, these three threads leave an identifiable gap. Time series classification methods (Sect. 2.1) are compared almost exclusively via aggregate accuracy, obscuring instance-level fragility. Robustness auditing methods (Sect. 2.2) characterize failure through region-level performance mapping or global sensitivity indices, operating at the level of the model or the dataset rather than at the level of an individual, semantically meaningful transition between classes. Dataset morphing techniques (Sect. 2.3) generate the controlled perturbations that could support such an instance-level view, but have to our knowledge been applied to algorithm benchmarking (Correia et al., 2019) or forecasting evaluation (Santos et al., 2024), not to auditing a classifier's decision boundary directly; nor, to our knowledge, has the morphing operator itself been validated for whether it preserves or recovers boundary position under interpolation. tsMIST closes this gap by combining sample-level borderline-pair selection with morphing sequence generation and a boundary-position metric whose behaviour we validate directly (Sect. 3.3.1), rather than assumed.

3 Proposed Approach

This section presents tsMIST, a morphing-based framework for evaluating classifier robustness in time series classification. We first establish formal definitions and background concepts. We then introduce our sample selection methodology based on DTW distance and define metrics for quantifying decision boundary stability under controlled temporal deformations.

3.1 Background and Definitions

Definition 1 (*Time Series*) Following (Mohammadi Foumani et al., 2024), a time series is defined as an ordered sequence $X = \{(x_1, t_1), \dots, (x_T, t_T)\}$ where $x_i \in \mathbb{R}^d$ represents observations at timestamps t_i . The series is univariate when $d = 1$ and multivariate otherwise.

Definition 2 (*Time Series Classification*) In supervised time series classification, we are given a labeled dataset $D = \{(X_i, Y_i)\}_{i=1}^N$ where each X_i is a time series and $Y_i \in \{1, \dots, C\}$ is a discrete class label. The objective is to learn a classifier $f : \mathbb{R}^{T \times d} \rightarrow \{1, \dots, C\}$ by finding parameters θ^* that minimize prediction error:

$$\theta^* = \underset{\theta}{\operatorname{argmin}} L(Y, f(\theta; X)),$$

where L is a loss function (e.g., cross-entropy) and $f(\theta; X)$ produces class predictions.

Definition 3 (*Dataset Morphing*) Originally introduced by Correia et al. (2019) for algorithm analysis, dataset morphing generates intermediate datasets through controlled transformations. Given a source dataset $D^{(s)}$ and target dataset $D^{(t)}$, the morphed sequence is defined as:

$$\mathcal{D}^{(morph)} = \{D_i \mid D_0 = D^{(s)}, D_n = D^{(t)}, D_i = \tau(D_{i-1})\}, \quad 1 \leq i < n,$$

where τ represents an incremental transformation operator.

Definition 4 (*Time Series Morphing*) Santos et al. (2024) adapted dataset morphing to time series through linear interpolation (tsMorph):

$$X_i = \alpha_i \cdot X^{(t)} + (1 - \alpha_i) \cdot X^{(s)},$$

where $\alpha_i \in [0, 1]$ is the morphing parameter, $X^{(s)}, X^{(t)} \in \mathbb{R}^{T \times d}$ are source and target series, and X_i is the interpolated series at step i (Fig. 1). Note that this operator is *affine* in α_i : every entry of X_i is a fixed linear combination of the corresponding entries of $X^{(s)}$ and $X^{(t)}$ at the *same* time index. This property is convenient: it makes the position of a scalar decision boundary along the path analytically recoverable (Proposition 1). It also means, however, that when the discriminative signal between classes is the *temporal location* of a feature

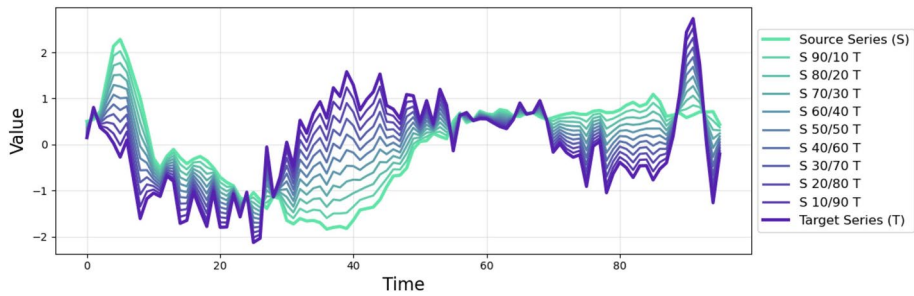


Fig. 1 Linear gradual morphing of the source time series into the target

rather than its amplitude, linear morphing does not relocate that feature: it instead blends the source and target amplitudes at both locations simultaneously, so a feature present in $X^{(s)}$ at position p_s and in $X^{(t)}$ at position p_t can appear as two concurrent, partially-weighted features rather than as one feature sliding from p_s to p_t .

Definition 5 (DTW Alignment Path) For a source series $X_d^{(s)}$ and target series $X_d^{(t)}$ of equal length T (per dimension d), the Dynamic Time Warping alignment path is the index correspondence

$$\pi_d = ((i_1, j_1), \dots, (i_K, j_K)),$$

obtained by backtracking the minimal-cost path through the recursive cost matrix

$$\gamma(i, j) = (X_{d,i}^{(s)} - X_{d,j}^{(t)})^2 + \min\{\gamma(i-1, j), \gamma(i, j-1), \gamma(i-1, j-1)\},$$

with $\gamma(0, 0) = 0$. By construction, π_d is monotone in both indices and satisfies the boundary conditions $i_1 = j_1 = 1$ and $i_K = j_K = T$.

Definition 6 (Path-Interpolation Morphing (PathI)) To preserve temporal structure while remaining faithful to the source and target endpoints, we interpolate the DTW correspondence π_d itself rather than the raw amplitude sequences. For a matched pair $(i, j) \in \pi_d$ and morphing weight $\alpha \in [0, 1]$, define the interpolated time position and amplitude

$$\tau_{(i,j)}(\alpha) = (1 - \alpha) i + \alpha j, \quad v_{(i,j)}(\alpha) = (1 - \alpha) X_{d,i}^{(s)} + \alpha X_{d,j}^{(t)}.$$

The morphed series at step α_i is obtained by resampling the scattered set $\{(\tau_{(i,j)}(\alpha_i), v_{(i,j)}(\alpha_i))\}_{(i,j) \in \pi_d}$ onto the regular grid $\{1, \dots, T\}$, averaging colliding positions and linearly interpolating gaps:

$$X_{i,d} = \rho\left(\{(\tau_{(i,j)}(\alpha_i), v_{(i,j)}(\alpha_i))\}_{(i,j) \in \pi_d}\right),$$

where $\rho(\cdot)$ denotes this resampling operator.

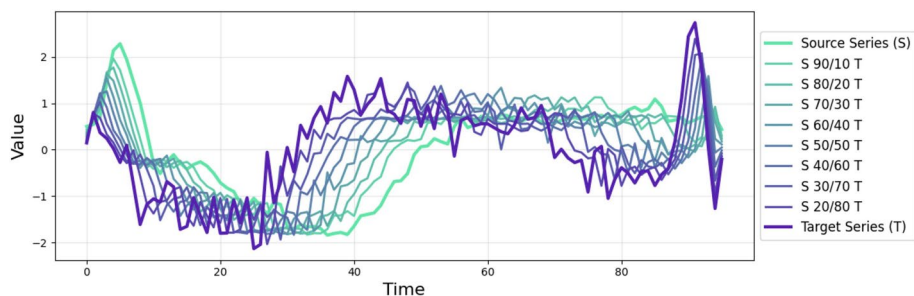


Fig. 2 PathI gradual morphing of the source time series into the target: matched features slide continuously from their source to their target position

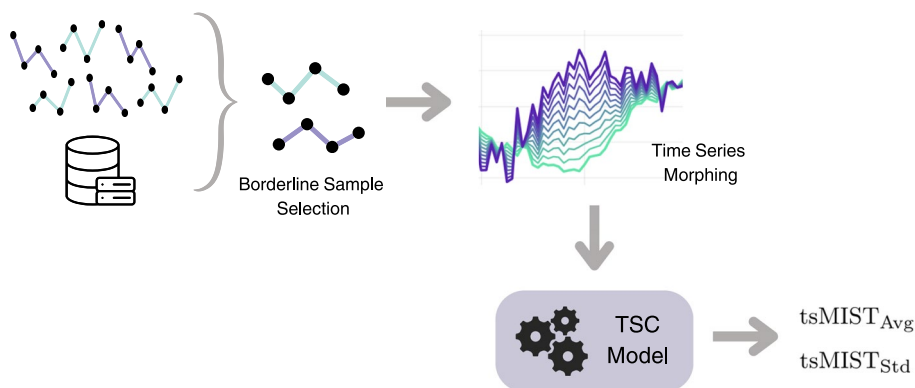


Fig. 3 tsMIST framework overview: time series data for classification task, borderline sample selection, morphing sequence generation, and sensitivity analysis

By the boundary conditions of π_d (Definition 5), $\tau_{(i,j)}(0) = i$ and $\tau_{(i,j)}(1) = j$ for every matched pair, so the operator recovers the endpoints *exactly*: $X_d|_{\alpha=0} = X_d^{(s)}$ and $X_d|_{\alpha=1} = X_d^{(t)}$. Unlike linear morphing (Definition 4, Fig. 1), which fixes time positions and blends amplitudes, PathI fixes the amplitude-time correspondence and displaces *position*: each matched feature slides continuously from its source location to its target location, so that morphology is preserved as a single evolving feature rather than a superposition of two (Fig. 2).

3.2 Model Sensitivity Analysis Framework

Our tsMIST framework (Fig. 3) extends time series morphing to classifier robustness analysis by operating at the sample level rather than the dataset level. The contribution of this framework is not either morphing operator in isolation (both linear interpolation and DTW-correspondence interpolation are established alignment techniques individually), but the sample-level borderline-pair protocol together with the boundary-position validation methodology (Sect. 3.3.1) that lets the two operators be compared quantitatively rather than

assumed. The methodology proceeds in three stages: borderline sample selection, morphing sequence generation, and sensitivity quantification.

3.2.1 Borderline Sample Selection

We identify challenging sample pairs, instances from different classes with high temporal similarity, as morphing candidates. These borderline cases are critical for probing decision boundary stability.

Definition 7 (*Candidate Pairs*) For a given class $c \in \{1, \dots, C\}$ and labeled dataset $D = \bigcup_{c=1}^C D_c$, the set of candidate pairs is as follows, where D_c contains all instances of class c :

$$P_c = \left\{ (X^{(s)}, X^{(t)}) \mid X^{(s)} \in D_c, X^{(t)} \in \bigcup_{c' \neq c} D_{c'} \right\}.$$

Definition 8 (*DTW Distance*) Dynamic Time Warping provides an elastic similarity measure between time series:

$$\text{DTW} : \mathbb{R}^{T \times d} \times \mathbb{R}^{T \times d} \rightarrow \mathbb{R}^+.$$

Definition 9 (*Morphing Candidates*) The final set of morphing pairs M is obtained by selecting the δ pairs with minimal DTW distance:

$$M = \underset{\substack{(X^{(s)}, X^{(t)}) \in P_c \\ |M| = \delta}}{\text{argmin}} \left\{ \text{DTW}(X^{(s)}, X^{(t)}) \right\},$$

where δ is a hyperparameter controlling the number of evaluation pairs. The choice of δ balances specificity and representativeness: low values focus exclusively on the most ambiguous cases, potentially underestimating robustness; high values may dilute the assessment by including less challenging pairs. We assess this trade-off empirically via a sensitivity analysis over a grid of δ values in Sect. 5.1.1, rather than fixing δ analytically. Conversely, for each pair $(X^{(s)}, X^{(t)}) \in M$ with labels $Y^{(s)} = c$ and $Y^{(t)} = c' \neq c$, we generate n intermediate series using Definition 4 and 6, creating a controlled transformation path from source to target.

3.2.2 Sensitivity Metrics

Definition 10 (*Switch Threshold*) For a classifier f and morphing sequence $\{X_i\}_{i=1}^n$ from source class c to target class c' , define

$$\text{switchThreshold} = \begin{cases} \min \left\{ \frac{i}{n} \mid f(X_i) \neq c, 1 \leq i \leq n \right\}, & \text{if } f(X_0) = c \text{ and } \exists i : f(X_i) \neq c \quad (\text{OK}) \\ \text{UNDEFINED}, & \text{if } f(X_0) \neq c \quad (\text{BASE_FAIL}) \\ \text{UNDEFINED}, & \text{if } f(X_0) = c \text{ and } f(X_i) = c \forall i \quad (\text{NO_SWITCH}) \end{cases}$$

representing the normalized morphing point where the classifier's prediction first deviates from the source class. Unlike prior treatments that collapse both undefined cases to 0, we keep `base_fail` (the model already misclassifies the unperturbed source) and `no_switch` (maximal rigidity: the model never reclassifies) as distinct, explicit outcomes: the former reflects a failure of the base classifier rather than of its sensitivity to morphing, and conflating it with fragility inverts the intended interpretation.

Definition 11 (*Coverage and Path Monotonicity*) For a set of δ morphing pairs, let $V \subseteq \{1, \dots, \delta\}$ index the pairs with an ok switch threshold (Definition 10). We define

$$\text{coverage} = \frac{|V|}{\delta},$$

the fraction of pairs yielding a defined switch threshold, and, for each $k \in V$, a monotonicity indicator equal to 1 if f does not revert to class c at any step after the first switch, and 0 otherwise. We report the mean monotonicity fraction alongside coverage as a diagnostic: `switchThreshold` is defined via the *first* crossing, so a non-monotone (oscillating) response causes it to underestimate the point of stable reclassification.

Definition 12 (*tsMIST Metrics*) Given δ morphing pairs, let $\{\text{switchThreshold}_k\}_{k \in V}$ be the subset with ok status (Definition 11). We define

$$\begin{aligned} \text{tsMIST}_{\text{Avg}} &= \frac{1}{|V|} \sum_{k \in V} \text{switchThreshold}_k, \\ \text{tsMIST}_{\text{Std}} &= \sqrt{\frac{1}{|V|} \sum_{k \in V} (\text{switchThreshold}_k - \text{tsMIST}_{\text{Avg}})^2}, \end{aligned}$$

computed only over V ; coverage (Definition 11) is reported alongside as a first-class output rather than folded into the average, so that a low-coverage regime (many undefined pairs) is visible instead of being masked by a numerically low but misleadingly-computed $\text{tsMIST}_{\text{Avg}}$.

3.3 Interpretation

Proposition 1 (*Boundary-Position Recovery*) Let $\phi : \mathbb{R}^{T \times d} \rightarrow \mathbb{R}$ be a linear functional of the series (e.g., the sample mean along a fixed dimension) and let f respond monotonically to ϕ with decision boundary ϕ_{bd} , i.e., $f(X) = c'$ iff $\phi(X) > \phi_{bd}$ (analogously for $<$). Under linear morphing (Definition 4), $\phi(X_\alpha) = (1 - \alpha)\phi(X^{(s)}) + \alpha\phi(X^{(t)})$ by linearity of ϕ , so the switch occurs exactly at

$$\alpha^* = \frac{\phi_{bd} - \phi(X^{(s)})}{\phi(X^{(t)}) - \phi(X^{(s)})},$$

and, in the continuous limit ($n \rightarrow \infty$), $\text{switchThreshold} \rightarrow \alpha^*$. Averaged over pairs, $\text{tsMIST}_{\text{Avg}} \rightarrow \mathbb{E}[\alpha^*]$: the metric is an estimator of decision-boundary position along the morphing path.

Corollary 1 (0.5 as a Symmetry Reference, Not a Universal Ideal) *If the boundary lies at the midpoint of the pair, $\phi_{bd} = (\phi(X^{(s)}) + \phi(X^{(t)}))/2$, then $\alpha^* = 0.5$ by Proposition 1. This is the symmetric case. Because the DTW-nearest pair selection of Definition 9 does not enforce this symmetry, $\alpha^* = 0.5$ is not guaranteed even under the idealized conditions of Proposition 1, and a value of $\text{tsMIST}_{\text{Avg}}$ away from 0.5 does not by itself indicate a poorly-calibrated or non-robust classifier.*

Under the linear- ϕ , monotone-response assumptions of Proposition 1, this motivates reading $\text{tsMIST}_{\text{Avg}}$ as a boundary-position estimate whose deviation from 0.5 reflects directional bias (rigidity toward the source class if $\text{tsMIST}_{\text{Avg}} > 0.5$, fragility if < 0.5) rather than distance from an ideal. We adopt this as an interpretive lens for the results that follow; we do not claim that the classifiers evaluated in this work satisfy linearity or monotonicity exactly, and the lens should be read accordingly, as a guide to interpretation rather than a property established for each model.

Remark 1 (Scope and Resolution) Proposition 1 assumes a linear ϕ and a monotone response; it formally applies to linear morphing and to amplitude-discriminative features, and its conclusions, including Corollary 1, should be understood as characterizing an idealized regime rather than as an empirically established property of the classifiers studied here. PathI morphing (Definition 6) is not affine in α in the same sense: by design, it displaces feature position rather than blending amplitude at a fixed position, so the closed-form α^* above does not transfer to it, and we do not currently have an analogous formal guarantee for that operator. Whether either operator recovers boundary position in practice, for classifiers that need not satisfy linearity or monotonicity, is an empirical question that this framework makes precisely statable but that we leave open here. Independently of the morphing operator, discretizing α over n steps limits switchThreshold 's resolution to $1/n$ and induces a small upward bias (of order $1/2n$) from snapping the true crossing to the next grid point above it; we report n alongside results for this reason.

The tsMIST metrics provide complementary insights into classifier robustness:

- $\text{tsMIST}_{\text{Avg}}$: Decision Boundary Position: Values near the pair's symmetric reference (nominally 0.5, per Corollary 1) indicate a boundary positioned midway between source and target.
- Low values indicate a boundary shifted toward the source, i.e., fragility, since little morphing is required to trigger reclassification.
- High values indicate a boundary shifted toward the target, i.e., rigidity, with the model resisting reclassification despite substantial transformation toward the target class.
- $\text{tsMIST}_{\text{Std}}$: Consistency Across Feature Space: Low standard deviation reflects uniform boundary position across different borderline pairs.

- High standard deviation reveals region-dependent stability, identifying areas where the decision surface is particularly vulnerable.

The synthetic setup used here is deliberately simpler and more controlled than the sine-wave experiments of Sect. 5.1: its role is to isolate a property of the morphing operator against a classifier with a known, fixed boundary, not to characterize the behaviour of the trained classifiers evaluated later in this work.

3.3.1 Synthetic Validation of Boundary-Position Recovery

Proposition 1 is exact for a linear functional ϕ under linear morphing, but it says nothing, by construction, about (i) whether PathI morphing recovers boundary position when ϕ is *not* linear, or (ii) how either operator behaves when the discriminative signal is not an amplitude feature but the *temporal location* of a feature, which is the regime PathI morphing (Definition 6) was designed for. We address both with a controlled synthetic experiment using an analyzable classifier whose decision boundary is fixed by construction, isolating the morphing operator from any classifier-fitting effects.

We consider two synthetic axes. On the *offset axis*, source and target series differ by a constant vertical offset $\phi(X) = \text{mean}(X)$, and the oracle classifier thresholds ϕ directly (the regime for which Proposition 1 was derived). On the *peak axis*, source and target series each contain a single Gaussian bump whose *location* differs between classes, and the oracle classifier thresholds $\text{argmax}(X)$, a non-linear functional to which Proposition 1 does not apply. For each axis, we impose a boundary position p by placing the pair's endpoints asymmetrically around the oracle's known boundary (analogously to Definition 9, but with p swept on a grid rather than left to selection), morph with both operators, and measure $\text{tsMIST}_{\text{Avg}}$ against the imposed p .

Figure 4 reports the result. On the offset axis, both operators recover p with slope 1.00 and $R^2 = 1.00$ (linear: intercept +0.007; PathI: intercept +0.008), consistent with Proposition 1 and confirming that neither morphing operator privileges 0.5 as a value in itself: the metric tracks whatever boundary position the pair happens to have, and 0.5 emerges only for symmetric pairs, as Corollary 1 states. On the peak axis, the two operators diverge sharply: linear morphing reads $\text{tsMIST}_{\text{Avg}} \approx 0.50$ *regardless of the imposed p* (slope 0.00, $R^2 = 0.32$), while PathI recovers p almost exactly (slope 1.00, $R^2 = 1.00$). The mechanism is direct: linearly blending two amplitude-matched, non-overlapping bumps produces a crossfade whose dominant peak switches at $\alpha \approx 0.5$ irrespective of where the bumps are located, so linear morphing is structurally blind to boundary position whenever the discriminative signal is temporal location rather than amplitude; PathI instead slides the matched feature continuously from its source to its target position, and the argmax crosses the true boundary at the correct α .

This result should be read at the scope stated in it: it isolates the morphing operator under a classifier with a known, fixed boundary, and thereby demonstrates that path-interpolation's advantage on position-discriminative features is not merely qualitative motivation but a mechanistically explained and numerically verified property of the operator itself. Whether this advantage persists once the boundary is learned rather than given remains open, for the reasons discussed in the Remark above.

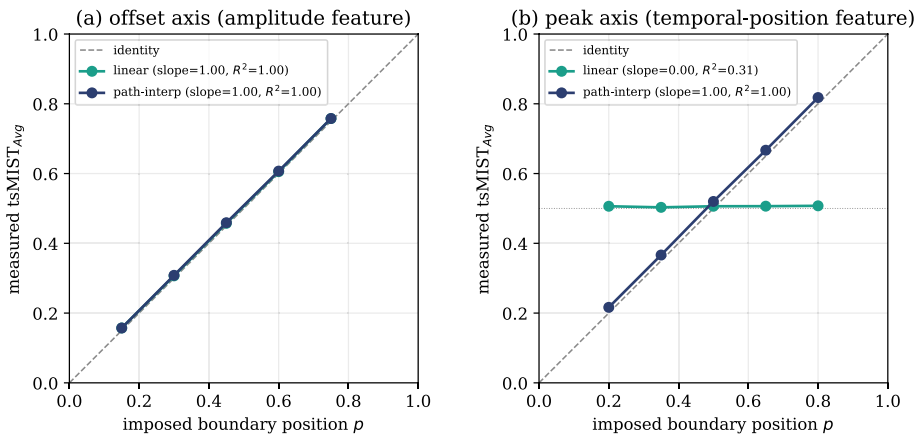


Fig. 4 Boundary-position recovery under an analyzable oracle classifier, for linear vs. PathI morphing. (a) Offset axis (amplitude-discriminative feature): both operators recover the imposed boundary position with $R^2 = 1.00$, confirming Proposition 1 and that 0.5 is not a privileged value per se. (b) Peak axis (position-discriminative feature): linear morphing is uninformative about boundary position (flat at ≈ 0.5), while PathI recovers it almost exactly, the controlled counterpart to the qualitative motivation given in Definition 6

Remark 2 (Why the First Two Moments) We do not claim that $\text{tsMIST}_{\text{Avg}}$ and $\text{tsMIST}_{\text{Std}}$ are sufficient statistics for the distribution of switchThreshold values across borderline pairs, in the formal sense of fully characterizing that distribution; no such claim would be warranted without a parametric assumption (e.g., normality) that we do not impose and that switchThreshold, being bounded in $[0, 1]$ and subject to the BASE_FAIL/NO_SWITCH exclusions of Definition 10, need not satisfy. Rather, the two statistics are chosen because they separate two distinct, independently interpretable properties: $\text{tsMIST}_{\text{Avg}}$ estimates the *location* of the switch threshold, tied under the idealized regime of Proposition 1 to decision-boundary position; $\text{tsMIST}_{\text{Std}}$ estimates its *spread*, indicating whether that position is stable or varies across the sampled pairs.

This decomposition can obscure a within-dataset distribution that is multimodal: a classifier highly fragile for one subset of borderline pairs and highly rigid for another could produce a middling $\text{tsMIST}_{\text{Avg}}$ and a large $\text{tsMIST}_{\text{Std}}$ that reads as generic inconsistency without revealing this bimodal structure. We partially mitigate this at a coarser grain by reporting $\text{tsMIST}_{\text{Avg}}$ and $\text{tsMIST}_{\text{Std}}$ themselves as distributions across datasets and models (Fig. 7), which exposes *between*-dataset heterogeneity; we do not, however, visualize the *within*-pair-set distribution of individual switchThreshold _{k} values, and a bimodal within-dataset regime would not be distinguishable, under the reporting adopted here, from a unimodal one sharing the same mean and standard deviation. We flag this as an acknowledged limitation of the two-moment summary rather than one this work fully resolves.

Unlike adversarial perturbation methods, which optimize a bounded perturbation of a *single* instance to induce misclassification, our morphing operators transform between *two real instances of different classes*, preserving temporal structure while systematically introducing features of the target class. This generates semantically meaningful variations that

models may encounter in real-world scenarios, making tsMIST analysis both interpretable and practically relevant for assessing classifier robustness.

4 Experiments

This section describes the experimental configuration used to validate tsMIST. We categorize our evaluation into two primary experimental tracks: controlled testing using synthetic data and practical validation using real-world medical time series. These components establish the basis for the results presented in Sect. 5.

Classifiers and Evaluation: For both synthetic and real-world tracks, we evaluated four distinct paradigms to observe how they respond to controlled waveform deformations. We use a simple RNN consisting of two LSTM layers (32 and 16 neurons) followed by dropout (0.2) and trained for 30 epochs. We also implemented a feature-based ensemble using the Catch22 (Lubba et al., 2019) feature set to train a Random Forest classifier. We used the ROCKET model (Dempster et al., 2020) with 5,000 random kernels and a Ridge classifier. These models were analyzed to determine how varying degrees of shifts affect decision boundaries. Finally, the InceptionTime classifier (Ismail Fawaz et al., 2020) was tested. This model utilizes Inception modules with bottleneck layers and multiple filter lengths to extract patterns at varying scales. To ensure convergence on these complex signals, the network was trained for 100 epochs with a batch size of 64.

4.1 Synthetic Data Experiments

To establish a baseline for tsMIST, we first utilized synthetically generated datasets. This environment allowed for controlled testing where the underlying structural differences between classes were known and adjustable.

Dataset Generation: The synthetic framework creates labeled time-series datasets with distinct patterns for two balanced classes ($n = 500$, shape: [500, 1, 100]). The process begins by generating basic waveforms, such as sine and cosine waves, which serve as the foundation for each class. To simulate real-world variability, Gaussian noise ($\mu = 0$) is added with an adjustable standard deviation. Additionally, vertical shifts are applied to alter amplitude, while horizontal shifts introduce time delays (phase shifts). The structural differences between classes resulting from these transformations are illustrated in Fig. 5.

4.2 Real-World Data Experiments

Following synthetic validation, we applied tsMIST to a diverse set of real-world physiological signals to ensure the framework's practical applicability and reliability. These experiments were run over 5 different seeds, the results were averaged with the corresponding variance.

Datasets: We evaluated tsMIST on 10 publicly available medical time series datasets from the Aeon archive, as summarized in Table 6. These datasets include electrocardiograms (ECG), electroencephalograms (EEG), and electrooculograms (EOG). The collection covers both univariate and multivariate recordings, with signal lengths ranging from

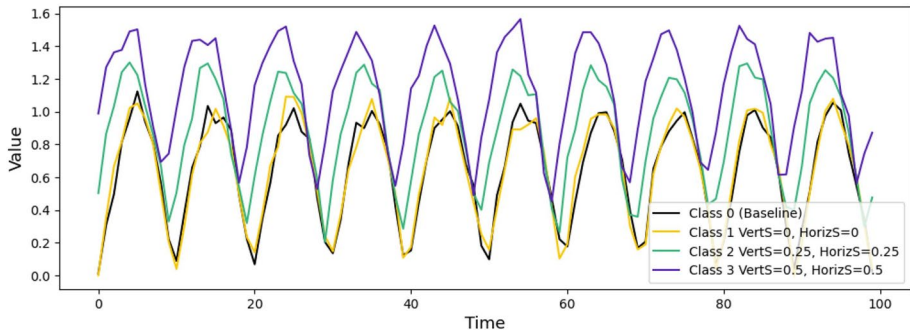


Fig. 5 Classes generated from sine waves with different vertical shifts (vertical and horizontal shifts)

82 to 2500 time steps and classification tasks involving 2 to 10 classes. All datasets were processed using an 80/20 train-test split.

Synthetic Data Auditing: To assess the realism of the synthetic transitions generated during the morphing process, we employed the pyMDMA framework (Façoco et al., 2025). This auditing tool allowed us to evaluate both the linear and PathI morphing approaches. We specifically analyzed the spectral coherence to assess frequency-domain preservation and the DTW distance to quantify the temporal alignment fidelity between the original series and the morphed synthetic outputs.

Explainability with TSFEL: To provide deeper interpretability for tsMIST results, we leveraged TSFEL (Time Series Feature Extraction Library)(Barandas et al., 2020) to compute the feature-wise differences between the Source series and the *switchThreshold* series. This approach enables interpretability at the meta-feature level, identifying which features changed the most at the *switchThreshold*. By analyzing these differences, we gain insights into the specific aspects of the time series that have the most impact on the model's predictions.

4.3 Implementation Details

The implementation of tsMIST is designed for both binary and multiclass tasks. For multiclass problems, the system identifies borderline pairs by comparing a user-specified target class against an aggregate of all remaining classes.

Data Handling: Input data follows the shape $[n_{samples}, n_{dimensions}, n_{timepoints}]$. Borderline pairs are identified following Definition 9, separating the data into target and source classes.

DTW Distance Calculation: Distances between samples are computed using the `aeon.distances.dtw_distance` function. For multivariate time series, the distances are averaged across all dimensions to produce a single distance metric per pair of samples.

Morphing Process: The `Tsmorph` class generates intermediate sequences where the granularity (default: 100) determines the number of steps between the source and target. In these experiments, we specifically contrast traditional linear interpolation with our proposed PathI morphing to determine which method better preserves signal morphology.

5 Results

This section presents the experimental findings. We first analyze tsMIST performance on synthetic datasets with varying class separation. We then examine results across 10 medical time series datasets. We conclude with synthetic data quality analysis, pertaining to the generated morphed series, and lastly with a meta-knowledge interpretation using TSFEL features to identify which signal characteristics drive prediction switches. These results demonstrate tsMIST's ability to detect stability patterns across different data types and model architectures.

5.1 Synthetic Data

Regarding the synthetic data experiments, we analyzed the influence of varying class-shift magnitudes on $tsMIST$ values. Figures 6a and 6b illustrate the performance of four architectures: LSTM, a Random Forest trained on Catch22 features, a Ridge Classifier trained on ROCKET kernels, and the InceptionTime classifier. The classification task is intentionally designed to be progressively simpler; by systematically increasing the shift between classes, we enhance their distinctness.

Figure 6a presents the robustness results using the linear morphing approach defined in Definition 4. The results indicate that InceptionTime is the most robust model overall, followed by LSTM, ROCKET, and Catch22. All models exhibit a downward trend in

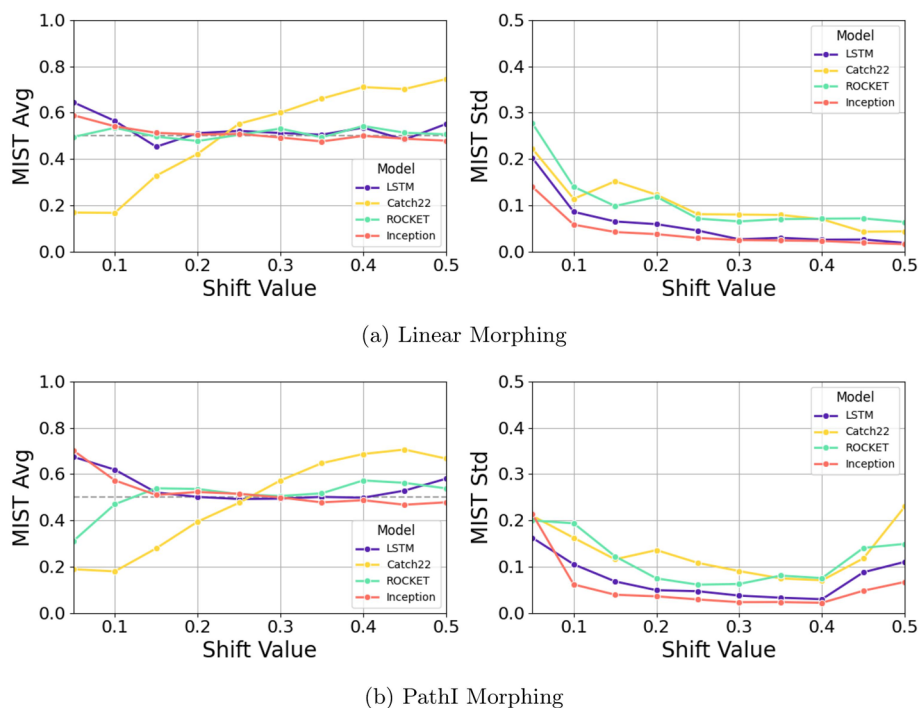


Fig. 6 Synthetic results for both **linear** and **PathI** morphing for the different models. $tsMIST_{Avg}$ on the left and $tsMIST_{Std}$ on the right. The dashed line at 50% $tsMIST_{Avg}$ marks the reference value expected under a symmetric source/target pair (Corollary 1), not a universal target for robustness

$tsMIST_{Std}$ as shift increases, suggesting that greater class separation enhances model certainty and reduces predictive variance. Notably, while all four models start with comparably high $tsMIST_{Std}$ (approximately 0.15–0.28) at minimal class separation, InceptionTime’s variance falls off fastest and settles near zero from moderate shift values onward, ending the range with both the lowest $tsMIST_{Std}$ and a $tsMIST_{Avg}$ close to the 50% reference. LSTM follows a similar, albeit slightly less stable, trajectory, converging to a low but non-negligible $tsMIST_{Std}$ (approximately 0.03–0.05).

In contrast, Catch22 proves more susceptible to perturbations, with $tsMIST_{Avg}$ values around 20% in low-shift scenarios, indicating a fragile decision boundary that switches with only minimal morphing. However, it is increasing toward 70% under high separation. This is further highlighted by a spike in $tsMIST_{Std}$ around shift = 0.15, an instability not observed in the deeper architectures, which decline more smoothly. Catch22’s persistently low $tsMIST_{Avg}$ at small shifts reflects the model’s difficulty establishing a stable decision boundary when classes substantially overlap.

Evaluation via the PathI morphing approach (Fig. 6b) reveals increased instability compared to the linear method. While LSTM and ROCKET approach the 50% symmetric reference (Corollary 1) in the mid-range of shift values, their $tsMIST_{Std}$ follows a U-shaped pattern: it declines through moderate shifts but rises again at high separation, unlike the largely monotonic decline seen under linear morphing. InceptionTime remains the most robust model under PathI as well, maintaining the lowest $tsMIST_{Std}$ across almost the entire shift range, while Catch22 remains the poorest performer, where its $tsMIST_{Avg}$ stays low (approximately 0.20–0.30) through low-to-moderate shifts, reflecting a persistently fragile boundary, and its $tsMIST_{Std}$ climbs sharply at high shift values (up to approximately 0.20), the largest such increase among the four models see(Fig.8).

5.1.1 Sensitivity Analysis of δ Configuration

The parameter δ in the $tsMIST$ framework controls the number of borderline sample pairs (Definition 9) used to compute the final sensitivity metrics. As discussed in Sect. 3.2, this choice involves an inherent trade-off: small values of δ restrict the analysis to the most ambiguous, DTW-nearest pairs, which sharpens the focus on worst-case boundary behaviour but risks a noisy, high-variance estimate of $tsMIST_{Avg}$ and $tsMIST_{Std}$ due to the limited sample; larger values incorporate less challenging pairs, diluting this focus but yielding a more statistically stable estimate. To characterise this trade-off empirically, we evaluated the stability of the $tsMIST$ metrics across $\delta \in \{50, 100, 250, 500, 1000\}$ on a synthetic dataset of size $n=500$, using PathI morphing.

Table 1 illustrates the performance of four distinct models across these settings. Two observations are relevant to selecting δ . First, across $\delta \in \{50, 100, 250, 500\}$, the standard deviation of $tsMIST_{Avg}$ across seeds does not show a clear monotonic decline for any of the four models (e.g., Catch22 ranges between 0.245 and 0.279, Rocket between 0.186 and 0.226), indicating that, within the range of δ values not exceeding n , increasing the pair budget does not by itself reduce estimator variance. Second, comparing $\delta = 500$ against $\delta = 1000$, the metrics remain largely consistent for most models: LSTM’s mean and standard deviation change only marginally (0.556 $\rightarrow$ 0.542; 0.162 $\rightarrow$ 0.158), as do Catch22’s (0.591 $\rightarrow$ 0.614; 0.279 $\rightarrow$ 0.256) and Inception’s (0.421 $\rightarrow$ 0.440; 0.091 $\rightarrow$ 0.075). Rocket is the exception, showing a more noticeable shift in its mean (0.397 $\rightarrow$ 0.570) alongside a

Table 1 *tsMIST* metrics (Mean and Std) across different δ settings (number of pairs chosen to compute the final score) over synthetic data size $n=500$, using PathI Morphing

δ	LSTM		Catch22		Rocket		Incep	
	Mean	Std	Mean	Std	Mean	Std	Mean	Std
50	0.610	0.132	0.654	0.245	0.415	0.226	0.413	0.102
100	0.605	0.134	0.629	0.253	0.410	0.221	0.411	0.110
250	0.609	0.146	0.589	0.274	0.387	0.225	0.394	0.121
500	0.556	0.162	0.591	0.279	0.397	0.186	0.421	0.091
1000	0.542	0.158	0.614	0.256	0.570	0.182	0.440	0.075

comparatively stable standard deviation ($0.186 \rightarrow 0.182$); we flag this as a case warranting further scrutiny rather than treating it as evidence against our chosen configuration. Overall, doubling δ from 500 to 1000 does not yield a systematic or substantial improvement in metric stability across models, while it doubles the number of morphing sequences that must be generated and evaluated. We therefore adopt $\delta = n$ as our operating rule, which for this synthetic setting corresponds to $\delta = 500$: this choice captures most of the stability achieved at higher δ while avoiding the added computational cost of evaluating a pair budget beyond the dataset size.

5.2 Real World Data

We extended our analysis to real-world data to evaluate algorithmic sensitivity to gradual input perturbations. This study utilizes ten medical datasets from the Aeon Time Series Classification Archive. Comprehensive dataset characteristics are provided in Table 6. Tables 2 and 3 summarise the average *tsMIST* scores across all classes alongside model accuracy for linear and PathI morphing, respectively. Detailed class-specific results are available in Appendix Tables 7 and 8.

In several instances, such as the StandWalkJump (SWJ) and EyesOpenShut (EOS) datasets, at least one classifier failed to classify any morphing candidates or retained only a single surviving pair, leaving $tsMIST_{Std}$ undefined. These cases are denoted by dashes (–) in Tables 2 and 3, highlighting model fragility that standard accuracy metrics fail to capture, even when multiple seeds are run. This phenomenon occurs under two conditions: (1) the model cannot correctly classify any morphing candidates, preventing the initiation of the morphing process ($tsMIST_{Avg} = 0.0$; undefined $tsMIST_{Std}$), the extreme case here is LSTM on EOS, which fails under both morphing schemes; or (2) the model correctly classifies only a single candidate pair, leaving no population for standard deviation calculation ($tsMIST_{Avg} \neq 0.0$; undefined $tsMIST_{Std}$), e.g., Catch22 and Rocket on SWJ.

Across both morphing approaches, two distinct patterns emerge. First, datasets where all models exhibit uniform robustness (or lack thereof), and second, datasets where models display divergent behavioural signatures. The latter is particularly valuable for comparative analysis, as it reveals specific algorithmic strengths and weaknesses. For example, in the EyesOpenShut (EOS) dataset under PathI morphing (Table 3), only LSTM fails outright ($tsMIST_{Avg} = 0.00$); the other three classifiers achieve comparable accuracy (50–56%) yet show markedly different robustness profiles. Catch22 and Rocket sit close together ($tsMIST_{Avg} = 0.32$, $tsMIST_{Std} \approx 0.21$ – 0.25), while InceptionTime has a lower $tsMIST_{Avg}$ of 0.20 alongside the tightest spread of any model on this dataset ($tsMIST_{Std} = 0.06$), indicating a systematically, rather than intermittently, fragile deci-

Table 2 Linear Morphing results per dataset, averaged across classes: MIST Average (M_{Avg}), MIST Standard Deviation (M_{Std}), and Model Accuracy (Acc), each shown with the average of the individual per-class standard deviations as a subscript

DS	LSTM			Catch22		
	M_{Avg}	M_{Std}	Acc	M_{Avg}	M_{Std}	Acc
ECG	0.505 ± 0.07	0.124 ± 0.07	0.87 ± 0.02	0.482 ± 0.03	0.174 ± 0.02	0.82 ± 0.02
EPI	0.440 ± 0.05	0.141 ± 0.07	0.71 ± 0.04	0.394 ± 0.02	0.158 ± 0.01	0.98 ± 0.00
ECGFD	0.510 ± 0.07	0.117 ± 0.01	1.00 ± 0.00	0.486 ± 0.03	0.123 ± 0.02	0.99 ± 0.00
SWJ	0.222 ± 0.14	0.110 ± 0.07	0.53 ± 0.14	0.482 ± 0.16	—	0.47 ± 0.07
TECG	0.508 ± 0.09	0.152 ± 0.01	0.99 ± 0.01	0.470 ± 0.02	0.201 ± 0.01	0.98 ± 0.00
ND	0.136 ± 0.21	0.026 ± 0.04	0.40 ± 0.03	0.479 ± 0.03	0.134 ± 0.03	1.00 ± 0.00
MI	0.371 ± 0.17	0.154 ± 0.05	0.70 ± 0.01	0.383 ± 0.05	0.201 ± 0.02	0.77 ± 0.01
COL	0.018 ± 0.03	0.006 ± 0.01	0.41 ± 0.04	0.216 ± 0.07	0.079 ± 0.04	0.41 ± 0.03
EOS	—	—	0.55 ± 0.00	0.300 ± 0.07	0.204 ± 0.06	0.56 ± 0.04
TOE	0.509 ± 0.05	0.199 ± 0.02	0.76 ± 0.04	0.473 ± 0.02	0.200 ± 0.01	0.96 ± 0.01
DS	Rocket			Incep		
	M_{Avg}	M_{Std}	Acc	M_{Avg}	M_{Std}	Acc
ECG	0.510 ± 0.02	0.125 ± 0.03	0.90 ± 0.02	0.497 ± 0.06	0.172 ± 0.03	0.87 ± 0.03
EPI	0.464 ± 0.01	0.142 ± 0.01	0.99 ± 0.01	0.473 ± 0.06	0.174 ± 0.04	0.90 ± 0.02
ECGFD	0.510 ± 0.02	0.056 ± 0.01	1.00 ± 0.00	0.510 ± 0.02	0.077 ± 0.01	1.00 ± 0.00
SWJ	0.017 ± 0.04	—	0.17 ± 0.00	0.175 ± 0.24	0.007 ± 0.02	0.50 ± 0.12
TECG	0.477 ± 0.01	0.133 ± 0.00	1.00 ± 0.00	0.510 ± 0.01	0.100 ± 0.00	1.00 ± 0.00
ND	0.417 ± 0.00	0.056 ± 0.00	1.00 ± 0.00	0.136 ± 0.19	0.068 ± 0.10	0.36 ± 0.01
MI	0.389 ± 0.06	0.221 ± 0.03	0.79 ± 0.03	0.446 ± 0.03	0.190 ± 0.02	0.80 ± 0.01
COL	0.239 ± 0.11	0.044 ± 0.04	0.28 ± 0.05	0.093 ± 0.10	0.012 ± 0.02	0.39 ± 0.01
EOS	0.507 ± 0.06	0.237 ± 0.04	0.50 ± 0.06	0.102 ± 0.23	—	0.51 ± 0.04
TOE	0.510 ± 0.00	0.137 ± 0.01	0.95 ± 0.01	0.510 ± 0.03	0.187 ± 0.03	0.80 ± 0.03

M_Avg: Bold values indicate performance within ± 10 percentage points of the 50% mark [0.4–0.6]

M_Std: Nobold values were applied

Acc: Bold values denote the highest-performing model across LSTM, Catch22, Rocket, and Inception

sion boundary. This distinction between "fragile but variable" and "fragile but consistent" behaviour is invisible to accuracy alone (51% vs. 50% vs. 56%) and is exactly the kind of instance-level signal tsMIST is designed to expose.

5.3 Morphing Comparison

Both morphing strategies systematically differentiate robustness levels across the evaluated models. The linear transformation described in Definition 4 provides a stable, continuous morphing process that clearly distinguishes model performance, though the morphological realism of the resulting synthetic series is a point of contention. Conversely, the PathI approach proposed in this work (Definition 6) offers a more sensitive robustness metric by interpolating the DTW alignment path between source and target series to better approximate the underlying data distribution.

As evidenced by Table 4, where DTW represents the distance between original and synthetic series, and Spectral Coherence measures signal consistency, the PathI approach achieves a lower DTW distance than Linear morphing on every dataset, with a consistent

Table 3 PathI Morphing results per dataset, averaged across classes: MIST Average (M_{Avg}), MIST Standard Deviation (M_{Std}), and Model Accuracy (Acc), each shown with the average of the individual per-class standard deviations as a subscript

DS	LSTM			Catch22		
	M_{Avg}	M_{Std}	Acc	M_{Avg}	M_{Std}	Acc
ECG	0.472 $\pm$ 0.06	0.152 $\pm$ 0.04	0.87 $\pm$ 0.02	0.432 $\pm$ 0.04	0.191 $\pm$ 0.03	0.82 $\pm$ 0.02
EPI	0.200 $\pm$ 0.11	0.141 $\pm$ 0.07	0.71 $\pm$ 0.04	0.293 $\pm$ 0.01	0.184 $\pm$ 0.01	0.98 $\pm$ 0.00
ECGFD	0.500 $\pm$ 0.04	0.101 $\pm$ 0.01	1.00 $\pm$ 0.00	0.470 $\pm$ 0.02	0.128 $\pm$ 0.01	0.99 $\pm$ 0.00
SWJ	0.114 $\pm$ 0.10	0.062 $\pm$ 0.04	0.53 $\pm$ 0.14	0.120 $\pm$ 0.20	0.029 $\pm$ 0.06	0.47 $\pm$ 0.07
TECG	0.500 $\pm$ 0.10	0.134 $\pm$ 0.02	0.99 $\pm$ 0.01	0.443 $\pm$ 0.01	0.178 $\pm$ 0.01	0.98 $\pm$ 0.00
ND	0.023 $\pm$ 0.03	0.019 $\pm$ 0.03	0.40 $\pm$ 0.03	0.440 $\pm$ 0.01	0.124 $\pm$ 0.01	1.00 $\pm$ 0.00
MI	0.304 $\pm$ 0.15	0.161 $\pm$ 0.09	0.70 $\pm$ 0.01	0.368 $\pm$ 0.03	0.216 $\pm$ 0.02	0.77 $\pm$ 0.01
COL	0.021 $\pm$ 0.05	0.010 $\pm$ 0.02	0.41 $\pm$ 0.04	0.158 $\pm$ 0.08	0.102 $\pm$ 0.04	0.41 $\pm$ 0.03
EOS	—	—	0.55 $\pm$ 0.00	0.323 $\pm$ 0.07	0.245 $\pm$ 0.06	0.56 $\pm$ 0.04
TOE	0.462 $\pm$ 0.09	0.246 $\pm$ 0.04	0.76 $\pm$ 0.04	0.412 $\pm$ 0.03	0.205 $\pm$ 0.02	0.96 $\pm$ 0.01
DS	Rocket			Incep		
	M_{Avg}	M_{Std}	Acc	M_{Avg}	M_{Std}	Acc
ECG	0.490 $\pm$ 0.05	0.176 $\pm$ 0.06	0.90 $\pm$ 0.02	0.485 $\pm$ 0.04	0.225 $\pm$ 0.04	0.87 $\pm$ 0.03
EPI	0.335 $\pm$ 0.02	0.201 $\pm$ 0.01	0.99 $\pm$ 0.01	0.396 $\pm$ 0.07	0.188 $\pm$ 0.05	0.90 $\pm$ 0.02
ECGFD	0.500 $\pm$ 0.01	0.036 $\pm$ 0.00	1.00 $\pm$ 0.00	0.500 $\pm$ 0.02	0.050 $\pm$ 0.01	1.00 $\pm$ 0.00
SWJ	0.004 $\pm$ 0.01	—	0.17 $\pm$ 0.00	0.112 $\pm$ 0.08	0.015 $\pm$ 0.03	0.50 $\pm$ 0.12
TECG	0.500 $\pm$ 0.00	0.059 $\pm$ 0.00	1.00 $\pm$ 0.00	0.500 $\pm$ 0.01	0.097 $\pm$ 0.01	1.00 $\pm$ 0.00
ND	0.361 $\pm$ 0.00	0.098 $\pm$ 0.01	1.00 $\pm$ 0.00	0.084 $\pm$ 0.12	0.028 $\pm$ 0.04	0.36 $\pm$ 0.01
MI	0.298 $\pm$ 0.05	0.214 $\pm$ 0.04	0.79 $\pm$ 0.03	0.387 $\pm$ 0.04	0.227 $\pm$ 0.02	0.80 $\pm$ 0.01
COL	0.159 $\pm$ 0.06	0.053 $\pm$ 0.02	0.28 $\pm$ 0.05	0.052 $\pm$ 0.08	0.010 $\pm$ 0.01	0.39 $\pm$ 0.01
EOS	0.321 $\pm$ 0.13	0.209 $\pm$ 0.08	0.50 $\pm$ 0.06	0.197 $\pm$ 0.28	0.061 $\pm$ 0.10	0.51 $\pm$ 0.04
TOE	0.496 $\pm$ 0.01	0.174 $\pm$ 0.01	0.95 $\pm$ 0.01	0.442 $\pm$ 0.05	0.224 $\pm$ 0.02	0.80 $\pm$ 0.03

M_Avg: Bold values indicate performance within ± 10 percentage points of the 50% mark [0.4–0.6]

M_Std: Nobold values were applied

Acc: Bold values denote the highest-performing model across LSTM, Catch22, Rocket, and Inception

Table 4 DTW distance and spectral coherence metrics (PathI morphing and linear morphing)

Dataset	DTW Distance		Spectral Coherence	
	PathI	Linear	PathI	Linear
ECG	5.12	6.21	2.40×10^{-3}	2.05×10^{-3}
EPI	17.6	21.1	7.75×10^{-4}	8.25×10^{-4}
ECGFD	2.85	3.56	5.95×10^{-3}	4.15×10^{-3}
SWJ	11.9	13.6	7.00×10^{-4}	2.30×10^{-3}
TECG	1.90	2.41	9.50×10^{-3}	5.95×10^{-3}
ND	0.136	0.168	9.00×10^{-4}	8.33×10^{-4}
MI	4.18	5.10	3.82×10^{-3}	2.61×10^{-3}
COL	153	190	2.08×10^{-3}	1.78×10^{-3}
EOS	2780	3460	1.53×10^{-2}	9.90×10^{-3}
TOE	9.04	10.6	1.75×10^{-3}	1.35×10^{-3}

Values rounded to 3 significant digits

reduction of roughly 13–21%, indicating that PathI-generated series remain closer in temporal alignment to the source signal regardless of dataset. PathI achieves higher coherence than Linear on eight of the ten datasets, sometimes substantially so (e.g., TwoLeadECG: 9.50×10^{-3} vs. 5.95×10^{-3}), but is lower on Epilepsy (7.75×10^{-4} vs. 8.25×10^{-4}) and, more markedly, on StandWalkJump (7.00×10^{-4} vs. 2.30×10^{-3} , roughly a threefold reduction). This exception suggests that PathI's benefit in frequency-domain fidelity is not universal and may depend on dataset-specific properties, such as signal length or the number of DTW-matched feature pairs available for interpolation. Overall, these results indicate that the PathI method consistently produces synthetic time series with higher temporal fidelity to the original source data and, typically, though not uniformly, better preserves spectral characteristics.

Figure 7 illustrates the distribution of $tsMIST_{Avg}$ and $tsMIST_{Std}$ for both approaches across all datasets and models. Figure 7b shows visually wider $tsMIST_{Std}$ boxes for PathI than for Linear on some models; however, this visual impression does not correspond to a practically meaningful difference once dataset-model pairing is accounted for. We defer the substantive comparison of stability between the two operators to the Statistical Analysis subsection that follows, rather than characterizing it here.

5.3.1 Meta-knowledge Interpretation

Table 5 lists the features exhibiting the most significant shifts at the *switchThreshold* for the ECG200 (ECG) dataset using the Inception Time model. Under Linear morphing, the identification of features such as Mel-frequency Cepstral Coefficients (MFCC) and Linear Prediction Cepstral Coefficients (LPCC) alongside Spectrogram Mean coefficients suggests sensitivity to both the spectral envelope and resonant structures of the signal. Specifically, for Class 1, LPCC 0 ranks first, likely reflecting sensitivity to shifts in waveform morphology associated with pathological events, while for Class -1, MFCC 2 dominates alongside temporal features such as Median Absolute Difference, indicating that the model also relies on signal variability to distinguish this class.

Under PathI morphing, the picture shifts substantially. For both classes, the top-ranked features are almost exclusively Spectrogram Mean coefficients concentrated in a narrow, adjacent high-frequency band (45–50 Hz), with Class 1 showing markedly larger feature-value shifts than Class -1 at the same frequencies (e.g., Spectrogram Mean at 50.0 Hz: $\sim$

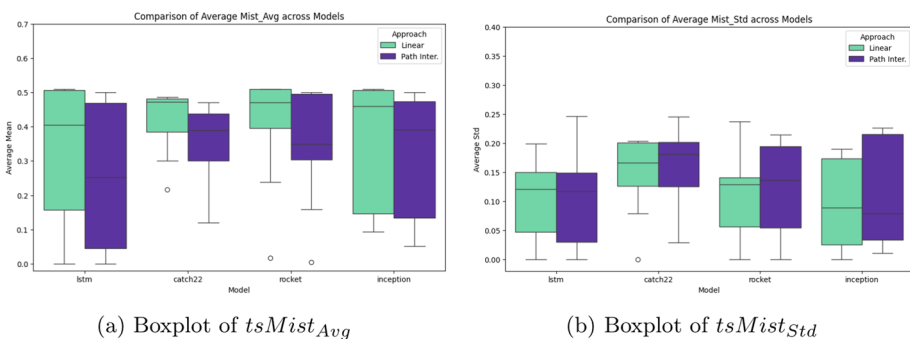


Fig. 7 Model results (lstm, catch, rocket, inception) across datasets using Linear Morphing (green) and PathI Morphing (purple), for all datasets

Table 5 Comparison of the top-5 most changed TSFEL features at the *switchThreshold* for the ECG200 dataset (Inception model) using Linear and PathI morphing

Rank	Linear morphing		PathI morphing	
	Class 1	Class -1	Class 1	Class -1
1	LPCC 0	MFCC 2	Spec Mean 50.0Hz	Spec Mean 50.0Hz
2	Spec Mean 41.94Hz	Median Abs Diff	Spec Mean 46.77Hz	Spec Mean 48.39Hz
3	Spec Mean 43.55Hz	Spec Mean 14.52Hz	Spec Mean 40.32Hz	LPCC 0
4	Spec Mean 45.16Hz	Spec Mean 19.35Hz	Spec Mean 48.39Hz	Mean
5	Spec Mean 37.1Hz	Spec Mean 25.81Hz	Spec Mean 45.16Hz	Spec Mean 46.77Hz

10,839 for Class 1 versus ~ 95.5 for Class -1). This is a departure from the Linear results: Linear morphing implicates a broader mix of feature families (LPCC, MFCC, Spectrogram Mean, and temporal-difference features) whereas the PathI-driven switch for this dataset and model is overwhelmingly explained by a cluster of neighbouring spectrogram frequency bins, with LPCC 0 and Mean appearing only in the lower-ranked PathI Class -1 positions and no MFCC feature appearing in either PathI column at all. This suggests that, for this dataset and model, PathI's temporal-position-preserving deformation concentrates the classifier's sensitivity onto a specific, localized spectral region rather than distributing it across the varied feature types implicated by Linear morphing.

The two morphing approaches probe these sensitivities differently: the linear approach clearly differentiates models but may produce less realistic series that focus primarily on amplitude and fixed frequency changes, while the PathI morphing approach maintains the inherent temporal structure of the signal through Dynamic Time Warping. This approach reveals vulnerabilities to non-linear temporal deformations, such as warped patterns and complex frequency shifts, that are often undetected by the linear baseline; in this case, that vulnerability manifests as a narrow-band spectral sensitivity rather than the broader cepstral and temporal sensitivity seen under Linear morphing. The PathI method also produces higher-fidelity synthetic series in general (Table 4); whether this comes at a cost in consistency $tsMIST_{Std}$ is addressed directly by the paired comparison in the following subsection of Statistical Analysis, rather than asserted here.

5.3.2 Statistical Analysis of Approach Relevance

To rigorously evaluate the performance of the PathI method against the established Linear baseline, we employed a Wilcoxon Signed-Rank test. This non-parametric test was chosen to compare the paired results across our test datasets, using raw $tsMIST_{Avg}$ and $tsMIST_{Std}$ values with no transformation applied. Grouping by dataset, model, and class yielded 144 paired observations in both the Linear and PathI conditions, all of which matched, so the test was conducted on the full set of comparable configurations with no groups excluded.

We treat these two metrics differently in interpretation. $tsMIST_{Avg}$ is a boundary-position estimate: a difference between operators indicates that Linear and PathI morphing locate the decision boundary differently along the morphing path, not that one operator is quantitatively superior to the other, since neither raw value is "better" in isolation. $tsMIST_{Std}$, by contrast, measures the consistency of the switch threshold across borderline pairs, where a lower value directly indicates a more consistent, less scattered decision boundary; here a lower-is-better comparison is meaningful.

The statistical analysis reveals that both metrics differ significantly between the Linear and PathI methods:

- Mean ($tsMIST_{Avg}$): We observed a highly statistically significant difference between the two methods ($p = 4.68 \times 10^{-9}$, $N = 144$), with a median difference of -0.0288 (PathI – Linear). This confirms that the two morphing operators are not interchangeable estimators of boundary position: they systematically disagree on where along the morphing path the switch occurs, consistent with their differing treatment of amplitude versus temporal-position features.
- Std. Deviation ($tsMIST_{Std}$): We also observed a statistically significant difference ($p = 0.0155$, $N = 144$), with PathI showing the lower value in the majority of pairs, as reflected in the rank-based test. Notably, the median difference across pairs was exactly 0.0000, meaning that for the typical configuration in our evaluation, the two operators produce the same $tsMIST_{Std}$; the significant result is driven by a consistent asymmetry in the minority of pairs where the two operators do differ, rather than by a uniformly small advantage spread across all pairs. We therefore do not characterize PathI as offering a practically meaningful improvement in boundary-position consistency over Linear morphing.

Taken together, these results establish that Linear and PathI morphing are statistically distinguishable operators rather than interchangeable approximations of the same quantity. This distinguishability is expected and consistent with Definitions 4 and 6: the two operators interpolate different quantities (amplitude at a fixed position versus feature position along the DTW correspondence) and are validated, in Sect. 3.3, as recovering boundary position under different assumptions about the discriminative signal. We therefore report the significant difference in $tsMIST_{Avg}$ as evidence that operator choice matters, without characterizing either operator's raw mean as superior, while the significant-but-median-null difference in $tsMIST_{Std}$ indicates the two operators are effectively at parity in the consistency of the boundaries they recover for most configurations.

6 Conclusion

This work introduced tsMIST, a framework for auditing time series classifier robustness through morphing between real, borderline instances of different classes. Its central contribution is not a new classification algorithm, nor either morphing operator considered in isolation, but an evaluation methodology whose sensitivity metric is formally and empirically validated rather than assumed: Proposition 1 establishes the conditions under which $tsMIST_{Avg}$ recovers true decision-boundary position, Corollary 1 clarifies that its commonly-assumed reference value of 0.5 reflects source–target symmetry rather than a universal target, and the synthetic validation of Sect. 3.3.1 confirms both results directly, on data with a known, fixed boundary.

Building on this validated foundation, we compared two morphing operators that instantiate the framework differently. Linear interpolation is affine in the morphing weight and inexpensive to compute, but blends source and target amplitude at a fixed time position; when a class's discriminative signal is the temporal location of a feature

rather than its amplitude, this blending can superimpose rather than relocate that feature, and our synthetic peak-axis experiment shows this makes linear morphing structurally blind to boundary position in that regime. PathI instead interpolates the DTW correspondence between source and target, displacing feature position directly; it lacks a closed-form guarantee analogous to Proposition 1 (Remark 2), and its reliance on a DTW alignment makes it more expensive to compute than linear interpolation, but it recovers boundary position exactly in the synthetic regime where linear morphing cannot.

On real-world data, this distinction is not merely synthetic. The two operators differ from each other on measures grounded in real physiological signals: PathI achieves lower DTW distance to the source signal on every evaluated dataset and higher spectral coherence on most, though this fidelity advantage is not universal across all ten datasets (Table 4). More directly relevant to the framework's purpose, a paired comparison across dataset-model configurations finds that the two operators systematically disagree on where the decision boundary lies along the morphing path (Sect. 5.3): this is a well-powered, directional effect, not noise, and is consistent with the position-versus-amplitude mechanism established synthetically. Evidence on relative stability, by contrast, is mixed: the synthetic experiments show a qualitatively different $tsMIST_{Std}$ pattern for PathI at high class separation, but the real-world paired comparison finds no practically meaningful difference in stability between the two operators once dataset-model pairing is accounted for. We report this contrast rather than resolving it in either direction, since the two experimental regimes probe different things: the synthetic setting isolates the morphing operator under maximal control, while the real-world comparison reflects the joint behaviour of operator and classifier together.

These results support a practical recommendation rather than a single preferred operator. Linear morphing is the lower-cost choice when a stable, easily-interpreted benchmark of amplitude-driven decision-surface behaviour is sufficient. PathI is the more appropriate choice when auditing robustness to temporally-localized, morphology-preserving deformations, particularly in domains such as clinical time series where the timing of a physiological event, and not merely its amplitude, may be the class-defining feature; this comes at the cost of greater computational expense and, currently, a less complete theoretical guarantee than the linear operator enjoys. We regard closing this theoretical gap for PathI, and extending the boundary-position validation of Sect. 3.3.1 to classifiers whose boundary is learned rather than given, as the most direct next steps for this line of work.

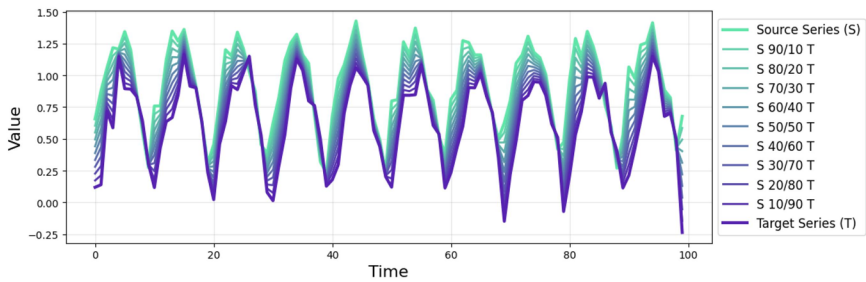
Appendix A

See Tables 6, 7, 8.

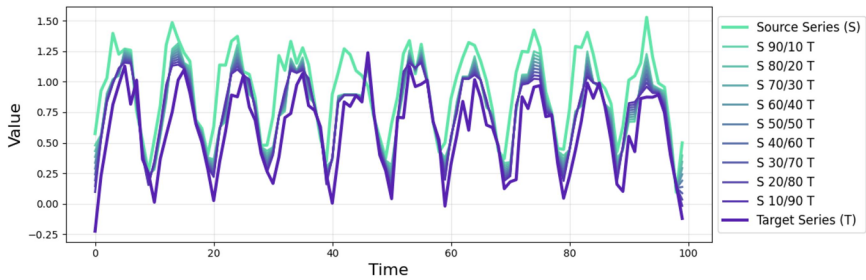
See Fig. 8.

Table 6 Dataset information, including training/testing set shapes and number of classes

Dataset	Train Shape	Test Shape	# Classes
Colposcopy (COL)	(128, 1, 180)	(32, 1, 180)	6
ECG200 (ECG)	(128, 1, 96)	(32, 1, 96)	2
ECGFiveDays (ECGFD)	(565, 1, 136)	(142, 1, 136)	2
Epilepsy (EPI)	(176, 3, 206)	(44, 3, 206)	4
EyesOpenShut (EOS)	(62, 14, 128)	(16, 14, 128)	2
MedicalImages (MI)	(729, 1, 99)	(183, 1, 99)	10
NerveDamage (ND)	(130, 1, 1500)	(33, 1, 1500)	3
StandWalkJump (SWJ)	(16, 4, 2500)	(5, 4, 2500)	3
ToeSegmentation1 (TOE)	(171, 1, 277)	(43, 1, 277)	2
TwoLeadECG (TECG)	(743, 1, 82)	(186, 1, 82)	2



(a) Linear Morphing



(b) PathI Morphing

Fig. 8 Morphing process of synthetic time series for baseline experiments

Table 7 Class-specific Linear Morphing results: MIST Average (*Avg*), MIST Standard Deviation (*Std*), and Model Accuracy (*Acc*), each reported as mean $\pm$ std across runs. Bold indicates *Avg* values between 0.45 and 0.55

Dataset	Class	%	LSTM			Catch22		
			<i>Avg</i>	<i>Std</i>	<i>Acc</i>	<i>Avg</i>	<i>Std</i>	<i>Acc</i>
ECG	-1	34	0.358 $\pm$ 0.060	0.122 $\pm$ 0.067	0.87 $\pm$ 0.02	0.330 $\pm$ 0.031	0.158 $\pm$ 0.024	0.82 $\pm$ 0.02
	1	66	0.651 $\pm$ 0.071	0.126 $\pm$ 0.074	0.87 $\pm$ 0.02	0.634 $\pm$ 0.033	0.191 $\pm$ 0.020	0.82 $\pm$ 0.02
EPI	Epil	24	0.611 $\pm$ 0.066	0.147 $\pm$ 0.090	0.71 $\pm$ 0.04	0.545 $\pm$ 0.020	0.154 $\pm$ 0.007	0.98 $\pm$ 0.00
	Run	26	0.365 $\pm$ 0.023	0.143 $\pm$ 0.062	0.71 $\pm$ 0.04	0.319 $\pm$ 0.023	0.152 $\pm$ 0.008	0.98 $\pm$ 0.00
	Saw	21	0.376 $\pm$ 0.059	0.136 $\pm$ 0.086	0.71 $\pm$ 0.04	0.380 $\pm$ 0.029	0.186 $\pm$ 0.023	0.98 $\pm$ 0.00
	Walk	28	0.410 $\pm$ 0.033	0.137 $\pm$ 0.046	0.71 $\pm$ 0.04	0.333 $\pm$ 0.006	0.141 $\pm$ 0.008	0.98 $\pm$ 0.00
ECGFD	1	50	0.443 $\pm$ 0.066	0.117 $\pm$ 0.009	1.00 $\pm$ 0.00	0.589 $\pm$ 0.043	0.139 $\pm$ 0.018	0.99 $\pm$ 0.00
	2	50	0.577 $\pm$ 0.066	0.117 $\pm$ 0.009	1.00 $\pm$ 0.00	0.383 $\pm$ 0.024	0.107 $\pm$ 0.016	0.99 $\pm$ 0.00
SWJ	Jump	29	0.286 $\pm$ 0.180	0.144 $\pm$ 0.084	0.53 $\pm$ 0.14	0.282 $\pm$ 0.160	-	0.47 $\pm$ 0.07
	Stand	33	0.380 $\pm$ 0.226	0.187 $\pm$ 0.139	0.53 $\pm$ 0.14	0.500 $\pm$ 0.285	0.47 $\pm$ 0.07	-
	Walk	38	-	-	0.53 $\pm$ 0.14	0.664 $\pm$ 0.027	-	0.47 $\pm$ 0.07
TECG	1	50	0.509 $\pm$ 0.097	0.151 $\pm$ 0.008	0.99 $\pm$ 0.01	0.285 $\pm$ 0.009	0.186 $\pm$ 0.006	0.98 $\pm$ 0.00
	2	50	0.506 $\pm$ 0.090	0.154 $\pm$ 0.014	0.99 $\pm$ 0.01	0.655 $\pm$ 0.023	0.216 $\pm$ 0.010	0.98 $\pm$ 0.00
ND	0	17	0.305 $\pm$ 0.438	0.039 $\pm$ 0.067	0.40 $\pm$ 0.03	0.753 $\pm$ 0.044	0.192 $\pm$ 0.069	1.00 $\pm$ 0.00
	1	36	0.103 $\pm$ 0.194	0.039 $\pm$ 0.067	0.40 $\pm$ 0.03	0.225 $\pm$ 0.018	0.147 $\pm$ 0.006	1.00 $\pm$ 0.00
	2	48	-	-	0.40 $\pm$ 0.03	0.459 $\pm$ 0.034	0.063 $\pm$ 0.015	1.00 $\pm$ 0.00
MI	1	10	0.529 $\pm$ 0.079	0.233 $\pm$ 0.041	0.70 $\pm$ 0.01	0.436 $\pm$ 0.022	0.227 $\pm$ 0.014	0.77 $\pm$ 0.01
	2	5	0.464 $\pm$ 0.111	0.227 $\pm$ 0.035	0.70 $\pm$ 0.01	0.333 $\pm$ 0.159	0.231 $\pm$ 0.035	0.77 $\pm$ 0.01
	3	7	0.437 $\pm$ 0.076	0.204 $\pm$ 0.030	0.70 $\pm$ 0.01	0.478 $\pm$ 0.042	0.200 $\pm$ 0.026	0.77 $\pm$ 0.01
	4	5	0.468 $\pm$ 0.428	0.070 $\pm$ 0.068	0.70 $\pm$ 0.01	0.494 $\pm$ 0.037	0.271 $\pm$ 0.015	0.77 $\pm$ 0.01
	5	4	0.412 $\pm$ 0.072	0.217 $\pm$ 0.036	0.70 $\pm$ 0.01	0.250 $\pm$ 0.027	0.180 $\pm$ 0.018	0.77 $\pm$ 0.01
	6	2	0.088 $\pm$ 0.196	0.017 $\pm$ 0.039	0.70 $\pm$ 0.01	0.350 $\pm$ 0.067	0.099 $\pm$ 0.018	0.77 $\pm$ 0.01
	7	4	0.435 $\pm$ 0.164	0.225 $\pm$ 0.044	0.70 $\pm$ 0.01	0.438 $\pm$ 0.046	0.273 $\pm$ 0.022	0.77 $\pm$ 0.01
	8	2	0.070 $\pm$ 0.156	0.029 $\pm$ 0.065	0.70 $\pm$ 0.01	0.276 $\pm$ 0.024	0.136 $\pm$ 0.006	0.77 $\pm$ 0.01
	9	10	0.383 $\pm$ 0.358	0.096 $\pm$ 0.107	0.70 $\pm$ 0.01	0.316 $\pm$ 0.064	0.168 $\pm$ 0.072	0.77 $\pm$ 0.01
	10	51	0.420 $\pm$ 0.094	0.219 $\pm$ 0.040	0.70 $\pm$ 0.01	0.455 $\pm$ 0.015	0.225 $\pm$ 0.016	0.77 $\pm$ 0.01
COL	0	9	-	-	0.41 $\pm$ 0.04	-	-	0.42 $\pm$ 0.03
	1	12	-	-	0.41 $\pm$ 0.04	0.004 $\pm$ 0.009	-	0.42 $\pm$ 0.03
	2	11	-	-	0.41 $\pm$ 0.04	0.444 $\pm$ 0.234	0.341 $\pm$ 0.045	0.42 $\pm$ 0.03
	3	25	-	-	0.41 $\pm$ 0.04	-	-	0.42 $\pm$ 0.03
	4	16	-	-	0.41 $\pm$ 0.04	0.760 $\pm$ 0.157	0.070 $\pm$ 0.157	0.42 $\pm$ 0.03
	5	28	0.109 $\pm$ 0.156	0.034 $\pm$ 0.049	0.41 $\pm$ 0.04	0.088 $\pm$ 0.033	0.063 $\pm$ 0.022	0.42 $\pm$ 0.03
EOS	0	55	-	-	0.55 $\pm$ 0.00	0.118 $\pm$ 0.061	0.079 $\pm$ 0.034	0.56 $\pm$ 0.04
	1	45	-	-	0.55 $\pm$ 0.00	0.482 $\pm$ 0.069	0.329 $\pm$ 0.095	0.56 $\pm$ 0.04
TOE	0	51	0.519 $\pm$ 0.054	0.200 $\pm$ 0.021	0.76 $\pm$ 0.04	0.474 $\pm$ 0.022	0.206 $\pm$ 0.016	0.96 $\pm$ 0.01
	1	49	0.499 $\pm$ 0.050	0.198 $\pm$ 0.021	0.76 $\pm$ 0.04	0.473 $\pm$ 0.016	0.193 $\pm$ 0.010	0.96 $\pm$ 0.01
Dataset	Class	%	Rocket			Incep		
			<i>Avg</i>	<i>Std</i>	<i>Acc</i>	<i>Avg</i>	<i>Std</i>	<i>Acc</i>
ECG	-1	34	0.402 $\pm$ 0.022	0.125 $\pm$ 0.034	0.90 $\pm$ 0.02	0.489 $\pm$ 0.081	0.167 $\pm$ 0.020	0.87 $\pm$ 0.03
	1	66	0.618 $\pm$ 0.022	0.125 $\pm$ 0.034	0.90 $\pm$ 0.02	0.505 $\pm$ 0.044	0.176 $\pm$ 0.039	0.87 $\pm$ 0.03
EPI	Epil	24	0.471 $\pm$ 0.015	0.172 $\pm$ 0.012	0.99 $\pm$ 0.01	0.718 $\pm$ 0.095	0.143 $\pm$ 0.037	0.90 $\pm$ 0.02
	Run	26	0.306 $\pm$ 0.006	0.081 $\pm$ 0.006	0.99 $\pm$ 0.01	0.533 $\pm$ 0.045	0.297 $\pm$ 0.042	0.90 $\pm$ 0.02
	Saw	21	0.539 $\pm$ 0.012	0.185 $\pm$ 0.007	0.99 $\pm$ 0.01	0.305 $\pm$ 0.099	0.145 $\pm$ 0.042	0.90 $\pm$ 0.02
	Walk	28	0.539 $\pm$ 0.019	0.129 $\pm$ 0.019	0.99 $\pm$ 0.01	0.336 $\pm$ 0.019	0.112 $\pm$ 0.036	0.90 $\pm$ 0.02
ECGFD	1	50	0.528 $\pm$ 0.023	0.056 $\pm$ 0.008	1.00 $\pm$ 0.00	0.565 $\pm$ 0.015	0.077 $\pm$ 0.009	1.00 $\pm$ 0.00
	2	50	0.492 $\pm$ 0.023	0.056 $\pm$ 0.008	1.00 $\pm$ 0.00	0.455 $\pm$ 0.015	0.077 $\pm$ 0.009	1.00 $\pm$ 0.00

Table 7 (continued)

Dataset	Class	%	Rocket			Incep		
			Avg	Std	Acc	Avg	Std	Acc
SWJ	Jump	29	—	—	0.17±0.00	0.178±0.398	-	0.50±0.12
	Stand	33	—	0.17±0.00	0.026±0.058	—	0.50±0.12	—
	Walk	38	0.052±0.116	—	0.17±0.00	0.321±0.250	0.021±0.047	0.50±0.12
TECG	1	50	0.522 ±0.017	0.139±0.003	1.00±0.00	0.556±0.014	0.100±0.002	1.00±0.00
	2	50	0.432±0.005	0.127±0.001	1.00±0.00	0.464 ±0.015	0.100±0.002	1.00±0.00
ND	0	17	0.582±0.002	0.071±0.002	1.00±0.00	0.213±0.293	0.102±0.149	0.36±0.01
	1	36	0.438±0.002	0.071±0.002	1.00±0.00	0.195±0.267	0.102±0.149	0.36±0.01
	2	48	0.232±0.002	0.024±0.001	1.00±0.00	-	-	0.36±0.01
MI	1	10	0.419±0.015	0.258±0.010	0.79±0.03	0.454 ±0.023	0.212±0.017	0.80±0.01
	2	5	0.444±0.019	0.174±0.069	0.79±0.03	0.383±0.053	0.196±0.017	0.80±0.01
	3	7	0.429±0.051	0.218±0.033	0.79±0.03	0.452 ±0.027	0.222±0.026	0.80±0.01
	4	5	0.444±0.040	0.244±0.015	0.79±0.03	0.465 ±0.009	0.159±0.007	0.80±0.01
	5	4	0.391±0.076	0.224±0.056	0.79±0.03	0.365±0.024	0.165±0.023	0.80±0.01
	6	2	0.334±0.079	0.171±0.028	0.79±0.03	0.465 ±0.020	0.083±0.014	0.80±0.01
	7	4	0.334±0.035	0.271±0.014	0.79±0.03	0.402±0.029	0.254±0.023	0.80±0.01
	8	2	0.268±0.022	0.192±0.018	0.79±0.03	0.280±0.020	0.167±0.006	0.80±0.01
	9	10	0.388±0.183	0.207±0.065	0.79±0.03	0.648±0.041	0.213±0.036	0.80±0.01
	10	51	0.436±0.071	0.255±0.015	0.79±0.03	0.541 ±0.045	0.228±0.019	0.80±0.01
COL	0	9	—	—	0.28±0.05	—	—	0.39±0.01
	1	12	0.292±0.296	0.026±0.058	0.28±0.05	—	—	0.39±0.01
	2	11	—	—	0.28±0.05	—	—	0.39±0.01
	3	25	0.201±0.195	0.050±0.074	0.28±0.05	-	-	0.39±0.01
	4	16	0.348±0.065	—	0.28±0.05	0.468 ±0.479	—	0.39±0.01
	5	28	0.593±0.087	0.186±0.100	0.28±0.05	0.090±0.100	0.069±0.103	0.39±0.01
EOS	0	55	0.452 ±0.054	0.237±0.044	0.50±0.06	0.018±0.040	—	0.51±0.04
	1	45	0.562±0.057	0.237±0.044	0.50±0.06	0.186±0.416	—	0.51±0.04
TOE	0	51	0.550 ±0.002	0.137±0.007	0.95±0.01	0.526 ±0.026	0.187±0.032	0.80±0.03
	1	49	0.470 ±0.002	0.137±0.007	0.95±0.01	0.494 ±0.026	0.187±0.032	0.80±0.03

M_Avg: Bold values indicate performance within ±5 percentage points of the 50% mark [0.45–0.55]

M_Std and Acc contain no bold formatting

Table 8 Class-specific pathi morphing results: MIST Average (*Avg*), MIST standard deviation (*Std*), and model accuracy (*Acc*), each reported as mean $\pm$ std across runs. **Bold** indicates *Avg* values between 0.45 and 0.55

Dataset	Class	%	LSTM			Catch22		
			<i>Avg</i>	<i>Std</i>	<i>Acc</i>	<i>Avg</i>	<i>Std</i>	<i>Acc</i>
ECG	-1	34	0.365 $\pm$ 0.055	0.140 $\pm$ 0.036	0.87 $\pm$ 0.02	0.310 $\pm$ 0.043	0.192 $\pm$ 0.019	0.82 $\pm$ 0.02
	1	66	0.578 $\pm$ 0.075	0.163 $\pm$ 0.046	0.87 $\pm$ 0.02	0.554 $\pm$ 0.044	0.189 $\pm$ 0.035	0.82 $\pm$ 0.02
EPI	Epil	24	0.198 $\pm$ 0.165	0.164 $\pm$ 0.121	0.71 $\pm$ 0.04	0.257 $\pm$ 0.015	0.189 $\pm$ 0.018	0.98 $\pm$ 0.00
	Run	26	0.194 $\pm$ 0.062	0.134 $\pm$ 0.034	0.71 $\pm$ 0.04	0.271 $\pm$ 0.005	0.149 $\pm$ 0.006	0.98 $\pm$ 0.00
	Saw	21	0.250 $\pm$ 0.123	0.135 $\pm$ 0.039	0.71 $\pm$ 0.04	0.455 $\pm$ 0.018	0.245 $\pm$ 0.023	0.98 $\pm$ 0.00
	Walk	28	0.159 $\pm$ 0.104	0.133 $\pm$ 0.093	0.71 $\pm$ 0.04	0.188 $\pm$ 0.007	0.152 $\pm$ 0.004	0.98 $\pm$ 0.00
ECGFD	1	50	0.438 $\pm$ 0.039	0.101 $\pm$ 0.012	1.00 $\pm$ 0.00	0.584 $\pm$ 0.020	0.129 $\pm$ 0.010	0.99 $\pm$ 0.00
	2	50	0.562 $\pm$ 0.039	0.101 $\pm$ 0.012	1.00 $\pm$ 0.00	0.356 $\pm$ 0.021	0.127 $\pm$ 0.005	0.99 $\pm$ 0.00
SWJ	Jump	29	0.126 $\pm$ 0.157	0.054 $\pm$ 0.051	0.53 $\pm$ 0.14	0.004 $\pm$ 0.009	—	0.47 $\pm$ 0.07
	Stand	33	0.216 $\pm$ 0.144	0.132 $\pm$ 0.077	0.53 $\pm$ 0.14	0.090 $\pm$ 0.201	—	0.47 $\pm$ 0.07
	Walk	38	—	—	0.53 $\pm$ 0.14	0.266 $\pm$ 0.397	0.086 $\pm$ 0.192	0.47 $\pm$ 0.07
TECG	1	50	0.495 $\pm$ 0.103	0.134 $\pm$ 0.017	0.99 $\pm$ 0.01	0.282 $\pm$ 0.013	0.167 $\pm$ 0.009	0.98 $\pm$ 0.00
	2	50	0.505 $\pm$ 0.103	0.134 $\pm$ 0.017	0.99 $\pm$ 0.01	0.604 $\pm$ 0.014	0.188 $\pm$ 0.008	0.98 $\pm$ 0.00
ND	0	17	—	—	0.40 $\pm$ 0.03	0.499 $\pm$ 0.009	0.132 $\pm$ 0.010	1.00 $\pm$ 0.00
	1	36	—	—	0.40 $\pm$ 0.03	0.457 $\pm$ 0.016	0.157 $\pm$ 0.015	1.00 $\pm$ 0.00
	2	48	0.069 $\pm$ 0.094	0.056 $\pm$ 0.077	0.40 $\pm$ 0.03	0.365 $\pm$ 0.015	0.082 $\pm$ 0.007	1.00 $\pm$ 0.00
MI	1	10	0.382 $\pm$ 0.217	0.194 $\pm$ 0.115	0.70 $\pm$ 0.01	0.494 $\pm$ 0.023	0.288 $\pm$ 0.011	0.77 $\pm$ 0.01
	2	5	0.493 $\pm$ 0.126	0.268 $\pm$ 0.061	0.70 $\pm$ 0.01	0.451 $\pm$ 0.077	0.172 $\pm$ 0.064	0.77 $\pm$ 0.01
	3	7	0.412 $\pm$ 0.050	0.250 $\pm$ 0.043	0.70 $\pm$ 0.01	0.398 $\pm$ 0.024	0.228 $\pm$ 0.008	0.77 $\pm$ 0.01
	4	5	0.439 $\pm$ 0.408	0.132 $\pm$ 0.127	0.70 $\pm$ 0.01	0.433 $\pm$ 0.028	0.280 $\pm$ 0.032	0.77 $\pm$ 0.01
	5	4	0.291 $\pm$ 0.117	0.203 $\pm$ 0.105	0.70 $\pm$ 0.01	0.219 $\pm$ 0.016	0.177 $\pm$ 0.018	0.77 $\pm$ 0.01
	6	2	0.050 $\pm$ 0.112	0.040 $\pm$ 0.090	0.70 $\pm$ 0.01	0.321 $\pm$ 0.015	0.133 $\pm$ 0.021	0.77 $\pm$ 0.01
	7	4	0.353 $\pm$ 0.062	0.224 $\pm$ 0.032	0.70 $\pm$ 0.01	0.459 $\pm$ 0.049	0.310 $\pm$ 0.026	0.77 $\pm$ 0.01
	8	2	0.050 $\pm$ 0.113	0.028 $\pm$ 0.062	0.70 $\pm$ 0.01	0.285 $\pm$ 0.014	0.162 $\pm$ 0.010	0.77 $\pm$ 0.01
	9	10	0.147 $\pm$ 0.202	0.098 $\pm$ 0.132	0.70 $\pm$ 0.01	0.273 $\pm$ 0.076	0.174 $\pm$ 0.033	0.77 $\pm$ 0.01
	10	51	0.421 $\pm$ 0.117	0.175 $\pm$ 0.100	0.70 $\pm$ 0.01	0.342 $\pm$ 0.018	0.236 $\pm$ 0.017	0.77 $\pm$ 0.01
COL	0	9	—	—	0.41 $\pm$ 0.04	0.155 $\pm$ 0.280	0.045 $\pm$ 0.101	0.42 $\pm$ 0.03
	1	12	—	—	0.41 $\pm$ 0.04	0.007 $\pm$ 0.006	0.001 $\pm$ 0.002	0.42 $\pm$ 0.03
	2	11	0.108 $\pm$ 0.241	0.053 $\pm$ 0.119	0.41 $\pm$ 0.04	0.260 $\pm$ 0.049	0.171 $\pm$ 0.024	0.42 $\pm$ 0.03
	3	25	—	—	0.41 $\pm$ 0.04	0.012 $\pm$ 0.002	0.002 $\pm$ 0.003	0.42 $\pm$ 0.03
	4	16	—	—	0.41 $\pm$ 0.04	0.481 $\pm$ 0.117	0.378 $\pm$ 0.097	0.42 $\pm$ 0.03
	5	28	0.018 $\pm$ 0.036	0.006 $\pm$ 0.014	0.41 $\pm$ 0.04	0.036 $\pm$ 0.015	0.015 $\pm$ 0.006	0.42 $\pm$ 0.03
EOS	0	55	—	—	0.55 $\pm$ 0.00	0.250 $\pm$ 0.057	0.203 $\pm$ 0.094	0.56 $\pm$ 0.04
	1	45	—	—	0.55 $\pm$ 0.00	0.395 $\pm$ 0.088	0.287 $\pm$ 0.024	0.56 $\pm$ 0.04
TOE	0	51	0.363 $\pm$ 0.077	0.237 $\pm$ 0.036	0.76 $\pm$ 0.04	0.452 $\pm$ 0.040	0.228 $\pm$ 0.020	0.96 $\pm$ 0.01
	1	49	0.560 $\pm$ 0.099	0.256 $\pm$ 0.043	0.76 $\pm$ 0.04	0.372 $\pm$ 0.027	0.181 $\pm$ 0.026	0.96 $\pm$ 0.01
Dataset	Class	%	Rocket			Incep		
			<i>Avg</i>	<i>Std</i>	<i>Acc</i>	<i>Avg</i>	<i>Std</i>	<i>Acc</i>
ECG	-1	34	0.474 $\pm$ 0.044	0.172 $\pm$ 0.060	0.90 $\pm$ 0.02	0.499 $\pm$ 0.037	0.226 $\pm$ 0.039	0.87 $\pm$ 0.03
	1	66	0.506 $\pm$ 0.052	0.181 $\pm$ 0.065	0.90 $\pm$ 0.02	0.471 $\pm$ 0.036	0.223 $\pm$ 0.037	0.87 $\pm$ 0.03
EPI	Epil	24	0.263 $\pm$ 0.022	0.197 $\pm$ 0.026	0.99 $\pm$ 0.01	0.610 $\pm$ 0.164	0.184 $\pm$ 0.086	0.90 $\pm$ 0.02
	Run	26	0.329 $\pm$ 0.036	0.203 $\pm$ 0.013	0.99 $\pm$ 0.01	0.509 $\pm$ 0.050	0.304 $\pm$ 0.026	0.90 $\pm$ 0.02
	Saw	21	0.546 $\pm$ 0.028	0.302 $\pm$ 0.009	0.99 $\pm$ 0.01	0.160 $\pm$ 0.012	0.122 $\pm$ 0.048	0.90 $\pm$ 0.02
	Walk	28	0.202 $\pm$ 0.005	0.102 $\pm$ 0.011	0.99 $\pm$ 0.01	0.304 $\pm$ 0.043	0.143 $\pm$ 0.025	0.90 $\pm$ 0.02
ECGFD	1	50	0.525 $\pm$ 0.013	0.036 $\pm$ 0.001	1.00 $\pm$ 0.00	0.506 $\pm$ 0.023	0.050 $\pm$ 0.007	1.00 $\pm$ 0.00
	2	50	0.475 $\pm$ 0.013	0.036 $\pm$ 0.001	1.00 $\pm$ 0.00	0.494 $\pm$ 0.023	0.050 $\pm$ 0.007	1.00 $\pm$ 0.00

Table 8 (continued)

Dataset	Class	%	Rocket			Incep		
			<i>Avg</i>	<i>Std</i>	<i>Acc</i>	<i>Avg</i>	<i>Std</i>	<i>Acc</i>
SWJ	Jump	29	—	—	0.17±0.00	0.008±0.018	—	0.50±0.12
	Stand	33	—	—	0.17±0.00	0.016±0.036	—	0.50±0.12
	Walk	38	0.012±0.022	—	0.17±0.00	0.312±0.195	0.044±0.098	0.50±0.12
TECG	1	50	0.509 ±0.003	0.059±0.002	1.00±0.00	0.494 ±0.006	0.097±0.007	1.00±0.00
	2	50	0.491 ±0.004	0.059±0.002	1.00±0.00	0.506 ±0.006	0.097±0.007	1.00±0.00
ND	0	17	0.371±0.006	0.108±0.010	1.00±0.00	0.234±0.339	0.075±0.112	0.36±0.01
	1	36	0.540 ±0.005	0.121±0.007	1.00±0.00	0.019±0.034	0.011±0.021	0.36±0.01
MI	2	48	0.173±0.003	0.065±0.008	1.00±0.00	—	—	0.36±0.01
	1	10	0.364±0.024	0.257±0.013	0.79±0.03	0.440±0.023	0.229±0.015	0.80±0.01
	2	5	0.473 ±0.076	0.251±0.066	0.79±0.03	0.305±0.089	0.228±0.039	0.80±0.01
	3	7	0.282±0.026	0.181±0.010	0.79±0.03	0.368±0.050	0.205±0.027	0.80±0.01
	4	5	0.340±0.049	0.255±0.021	0.79±0.03	0.521 ±0.017	0.247±0.008	0.80±0.01
	5	4	0.291±0.047	0.240±0.055	0.79±0.03	0.289±0.051	0.219±0.041	0.80±0.01
	6	2	0.120±0.068	0.098±0.063	0.79±0.03	0.342±0.031	0.220±0.017	0.80±0.01
	7	4	0.204±0.046	0.190±0.040	0.79±0.03	0.424±0.024	0.285±0.011	0.80±0.01
	8	2	0.202±0.036	0.163±0.025	0.79±0.03	0.257±0.010	0.162±0.006	0.80±0.01
	9	10	0.294±0.064	0.225±0.072	0.79±0.03	0.469 ±0.048	0.232±0.022	0.80±0.01
COL	10	51	0.409±0.022	0.284±0.007	0.79±0.03	0.461 ±0.029	0.238±0.027	0.80±0.01
	0	9	—	—	0.28±0.05	—	—	0.39±0.01
	1	12	0.068±0.045	0.046±0.043	0.28±0.05	—	—	0.39±0.01
	2	11	—	—	0.28±0.05	—	—	0.39±0.01
	3	25	0.288±0.219	0.020±0.045	0.28±0.05	0.002±0.004	—	0.39±0.01
EOS	4	16	0.341±0.052	0.017±0.033	0.28±0.05	0.178±0.398	—	0.39±0.01
	5	28	0.259±0.071	0.235±0.024	0.28±0.05	0.129±0.101	0.062±0.089	0.39±0.01
	0	55	0.195±0.180	0.127±0.090	0.50±0.06	0.108±0.162	0.063±0.102	0.51±0.04
	1	45	0.447±0.081	0.290±0.064	0.50±0.06	0.286±0.400	0.060±0.097	0.51±0.04
	0	51	0.511 ±0.009	0.175±0.013	0.95±0.01	0.290±0.046	0.202±0.024	0.80±0.03
TOE	1	49	0.482 ±0.008	0.172±0.014	0.95±0.01	0.595±0.052	0.245±0.013	0.80±0.03

M_Avg: Bold values indicate performance within ±5 percentage points of the 50% mark [0.45–0.55]

M_Std and Acc contain no bold formatting

Acknowledgements We would like to thank the Aeon Time Series Classification Repository for providing the open-access time series datasets used in this study. Google Gemini was used to support the formulation of English sentences based on author-provided input. All contents were carefully reviewed by the authors.

Author Contributions AB: Conceptualisation, Data curation, Formal analysis, Investigation, Methodology, Software, Visualisation, Writing — original draft, review and editing. CS: Conceptualisation, Funding acquisition, Validation, Supervision, Writing — review and editing. DF: Conceptualisation, Funding acquisition, Validation, Supervision, Writing — review and editing. MS: Conceptualisation, Validation, Supervision, Writing — review and editing.

Funding This work is a result of Agenda “Center for Responsible AI”, nr. C645008882-00000055, investment project nr. 62, financed by the Recovery and Resilience Plan (PRR) and by European Union - NextGeneration EU. Funded by the European Union – NextGenerationEU. Views and opinions expressed are however those of the author(s) only and do not necessarily reflect those of the European Union or the European Commission. Neither the European Union nor the European Commission can be held responsible for them. AISym4Med (101095387) supported by Horizon Europe Cluster 1: Health, ConnectedHealth (n.o 46858), supported by Competitiveness and Internationalisation Operational Programme (POCI) and Lisbon Regional Operational Programme (LISBOA 2020), under the PORTUGAL 2020 Partnership Agreement, through the European Regional Development Fund (ERDF). This work was financially supported by: UID/00027 of the LIACC - Artificial Intelligence and Computer Science Laboratory - funded by Fundação para a Ciência e a Tecnologia, I.P./MCTES through the national funds.

Data Availability No datasets were generated or analysed during the current study. The data used in this research is publicly available and can be downloaded from its official website: <https://www.timeseriesclassification.com/dataset.php>

Code Availability The code to reproduce the results presented in this paper is open-source and publicly available on GitHub: <https://github.com/Nia3324/tsMIST>

Declarations

Conflict of interest The authors have no conflict of interest to declare that are relevant to the content of this article.

Consent for publication All authors agreed with the content and that all gave explicit consent to submit and that they obtained consent from the responsible authorities at the institute/organization where the work has been carried out.

References

- Bagnall, A., Lines, J., Bostrom, A., Large, J., & Keogh, E. (2017). The great time series classification bake off: A review and experimental evaluation of recent algorithmic advances. *Data Mining and Knowledge Discovery*, 31, 606–660.
- Balendran, A., Beji, C., Bouvier, F., Khalifa, O., Evgeniou, T., Ravaud, P., & Porcher, R. (2025). A scoping review of robustness concepts for machine learning in healthcare. *npj Digital Medicine*, 8(1), 38.
- Barandas, M., Folgado, D., Fernandes, L., Santos, S., Abreu, M., Bota, P., Liu, H., Schultz, T., & Gamboa, H. (2020). Tsfel: Time series feature extraction library. *SoftwareX*, 11, 100456. <https://doi.org/10.1016/j.softx.2020.100456>
- Barandas, M., Folgado, D., Fernandes, L., Santos, S., Abreu, M., Bota, P., Liu, H., Schultz, T., & Gamboa, H. (2020). TSFEL: Time series feature extraction library. *SoftwareX*, 11, 100456. <https://doi.org/10.1016/j.softx.2020.100456>

- Belkhouja, T., Yan, Y., & Doppa, J. R. (2022). Dynamic time warping based adversarial framework for time-series domain. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 45(6), 7353–7366.
- Bellinger, C., Corizzo, R., & Japkowicz, N. (2021). Calibrated resampling for imbalanced and long-tails in deep learning. In *International conference on discovery science* Springer, (pp. 242–252).
- Brito, A., Santos, M., Folgado, D., & Soares, C. (2025). tsmist: Model sensitivity analysis with time series morphing. In *International conference on discovery science* (pp. 270–285). Springer.
- Cai, Y., Choudhry, A., Goswami, M., & Dubrawski, A. (2024). Timeseriesexam: A time series understanding exam [arXiv:2410.14752](https://arxiv.org/abs/2410.14752).
- Correia, A., Soares, C., & Jorge, A. (2019). Dataset morphing to analyze the performance of collaborative filtering. In *Discovery science: 22nd international conference, DS 2019, Split, Croatia, October 28–30, 2019, proceedings 22* (pp. 29–39). Springer.
- Dempster, A., Petitjean, F., & Webb, G. I. (2020). Rocket: Exceptionally fast and accurate time series classification using random convolutional kernels. *Data Mining and Knowledge Discovery*, 34(5), 1454–1495.
- Dong, J., Jiang, Z., Pan, D., Chen, Z., Guan, Q., Zhang, H., Gui, G., & Gui, W. (2025). A survey on confidence calibration of deep learning-based classification models under class imbalance data. *IEEE Transactions on Neural Networks and Learning Systems*, IEEE.
- Drapal, P., Prudêncio, R. B., & Silva Filho, T. M. (2024). Towards explainable evaluation: Explaining predicted performance using local performance regions. *Applied Soft Computing*, 167, 112351.
- Façoco, I. S., Rebelo, J., Matias, P., Bento, N., Morgado, A. C., Sampaio, A., Rosado, L., & Barandas, M. (2025). pymdma: Multimodal data metrics for auditing real and synthetic datasets. *SoftwareX*, 31, 102256. <https://doi.org/10.1016/j.softx.2025.102256>
- Garred, W., Oger, R., & Lauras, M. (2025). Evaluating the robustness of time series forecast models under disruptions. *IFAC-PapersOnLine*, 59(10), 829–834.
- Golling, T., Klein, S., Mastandrea, R., Nachman, B., & Raine, J. A. (2023). Morphing one dataset into another with maximum likelihood estimation. *Physical Review D*, 108(9), Article 096018.
- Inácio, R., Cerqueira, V., Barandas, M., & Soares, C. (2025). Meta-learning and data augmentation for stress testing forecasting models. In *International symposium on intelligent data analysis* (pp. 343–357). Springer.
- Ismail Fawaz, H., Forestier, G., Weber, J., Idoumghar, L., & Muller, P.-A. (2019). Deep learning for time series classification: A review. *Data Mining and Knowledge Discovery*, 33(4), 917–963.
- Ismail Fawaz, H., Lucas, B., Forestier, G., Pelletier, C., Schmidt, D. F., Weber, J., Webb, G. I., Idoumghar, L., Muller, P.-A., & Petitjean, F. (2020). Inceptiontime: Finding alexnet for time series classification. *Data Mining and Knowledge Discovery*, 34(6), 1936–1962.
- Ismail-Fawaz, A., Devanne, M., Berretti, S., Weber, J., & Forestier, G. (2025). Look into the lite in deep learning for time series classification. *International Journal of Data Science and Analytics*, 20(4), 4029–4049.
- Ismail-Fawaz, A., Devanne, M., Weber, J., & Forestier, G. (2022). Deep learning for time series classification using new hand-crafted convolution filters. In *2022 IEEE international conference on big data (Big Data)* (pp. 972–981). IEEE.
- Kielhöfer, L., Bosman, A. W., Hoos, H. H., & Rijn, J. N. (2025). Robustness margin: A new measure for the robustness of neural networks. In *International symposium on AI verification* (pp. 29–48). Springer.
- Kjærnlø, H. H., Mas-Ribas, L., Ashrafi, A., Sizov, G., Langseth, H., & Gundersen, O. E. (2024). Probing the robustness of time-series forecasting models with counterfactuals. [arXiv:2403.03508](https://arxiv.org/abs/2403.03508).
- Kurniawan, A., Putra, M. G., Hakim, D. L., & Ariyanto, M. (2026). Temporal adversarial attacks on time series and reinforcement learning systems: A systematic survey, taxonomy, and benchmarking roadmap.
- Lee, J.-G., Roh, Y., Song, H., & Whang, S.E. (2021). Machine learning robustness, fairness, and their convergence. In *Proceedings of the 27th ACM SIGKDD conference on knowledge discovery & data mining* (pp. 4046–4047).
- Lines, J., & Bagnall, A. (2015). Time series classification with ensembles of elastic distance measures. *Data Mining and Knowledge Discovery*, 29, 565–592.
- Liu, X., Aksu, T., Liu, J., Wen, Q., Liang, Y., Xiong, C., Savarese, S., Sahoo, D., Li, J., & Liu, C. (2025). Empowering time series analysis with synthetic data: A survey and outlook in the era of foundation models. [arXiv:2503.11411](https://arxiv.org/abs/2503.11411).

- Liu, X., Li, Y., He, Q., Zhu, J., Ji, W., Yao, A., & Zhu, J. (2026). Interp3d: Correspondence-aware interpolation for generative textured 3d morphing. [arXiv:2601.14103](https://arxiv.org/abs/2601.14103).
- Lo Piano, S., Sheikholeslami, R., Puy, A., & Saltelli, A. (2023). *Sensitivity auditing. The politics of modeling* (pp. 121–136).
- Lubba, C. H., Sethi, S. S., Knaute, P., Schultz, S. R., Fulcher, B. D., & Jones, N. S. (2019). catch22: Canonical time-series characteristics: Selected through highly comparative time-series analysis. *Data Mining and Knowledge Discovery*, 33(6), 1821–1852.
- Middlehurst, M., Schäfer, P., & Bagnall, A. (2024). Bake off redux: A review and experimental evaluation of recent time series classification algorithms. *Data Mining and Knowledge Discovery*, 38(4), 1958–2031.
- Mohammadi Foumani, N., Miller, L., Tan, C. W., Webb, G. I., Forestier, G., & Salehi, M. (2024). Deep learning for time series classification and extrinsic regression: A current survey. *ACM Computing Surveys*. <https://doi.org/10.1145/3649448>
- Muñoz, M. A., Villanova, L., Baatar, D., & Smith-Miles, K. (2018). Instance spaces for machine learning classification. *Machine Learning*, 107, 109–147.
- Neves, I., Folgado, D., Santos, S., Barandas, M., Campagner, A., Ronzio, L., Cabitza, F., & Gamboa, H. (2021). Interpretable heartbeat classification using local model-agnostic explanations on eegs. *Computers in Biology and Medicine*, 133, Article 104393.
- Nguyen, T. T., Le Nguyen, T., & Ifrim, G. (2024). Robust explainer recommendation for time series classification. *Data Mining and Knowledge Discovery*, 38(6), 3372.
- Pialla, G., Ismail Fawaz, H., Devanne, M., Weber, J., Idoumghar, L., Muller, P.-A., Bergmeir, C., Schmidt, D. F., Webb, G. I., & Forestier, G. (2025). Time series adversarial attacks: An investigation of smooth perturbations and defense approaches. *International Journal of Data Science and Analytics*, 19(1), 129–139.
- Resende, C., Folgado, D., Oliveira, J., Franco, B., Moreira, W., Oliveira-Jr, A., Cavaleiro, A., & Carvalho, R. (2021). Tip4. 0: Industrial internet of things platform for predictive maintenance. *Sensors*, 21(14), 4676.
- Rousseau, A.-J., Becker, T., Appeltans, S., Blaschko, M., & Valkenborg, D. (2025). Post hoc calibration of medical segmentation models. *Discover Applied Sciences*, 7(3), Article 180.
- Santos, M. R., Carvalho, A. C., & Soares, C. (2024). Enhancing algorithm performance understanding through tsmorph: generating semi-synthetic time series for robust forecasting evaluation. In *Proceedings*.
- Schäfer, P. (2015). The boss is concerned with time series classification in the presence of noise. *Data Mining and Knowledge Discovery*, 29, 1505–1530.
- Schardong, G., Novello, T., Paz, H., Medvedev, I., Da Silva, V., Velho, L., & Gonçalves, N. (2024). Neural implicit morphing of face images. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 7321–7330).
- Schwag, V., Bhagoji, A. N., Song, L., Sitawarin, C., Cullina, D., Chiang, M., & Mittal, P. (2019). Analyzing the robustness of open-world machine learning. In *Proceedings of the 12th ACM workshop on artificial intelligence and security* (pp. 105–116).
- Sousa, P., Soares, C., & Cerqueira, V. (2025). Evaluating predictive maintenance models in the presence of reflexivity: A case study in pharmaceutical manufacturing. In *Joint european conference on machine learning and knowledge discovery in databases* (pp. 333–343). Springer.
- Zhang, K., Zhou, Y., Xu, X., Dai, B., & Pan, X. (2024). Diffmorpher: Unleashing the capability of diffusion models for image morphing. In *CVPR*.

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.

Authors and Affiliations

Antónia Brito^{1,2} · Duarte Folgado^{3,4} · Carlos Soares^{1,2,3,5} · Moisés Santos^{1,2}

✉ Antónia Brito
antonia.c.brito2@gmail.com

Duarte Folgado
duarte.folgado@aicos.fraunhofer.pt

Carlos Soares
csoares@fe.up.pt

Moisés Santos
mrsantos@fe.up.pt

¹ Department of Informatics Engineering, Faculty of Engineering of the University of Porto, Rua Dr. Roberto Frias, 4200-465 Porto, Portugal

² Laboratory for Artificial Intelligence and Computer Science (LIACC), Rua Dr. Roberto Frias, 4200-465 Porto, Portugal

³ Fraunhofer Portugal AICOS, Rua Alfredo Allen, 455, 4200-135 Porto, Portugal

⁴ Comprehensive Health Research Centre (CHRC), Porto, Portugal

⁵ BrightFactory, Porto, Portugal